

THE UNIVERSITY OF CHICAGO

**How Crime Relates to Investment and Disinvestment in Residential
Properties: A Baltimore City Case Study**

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Abstract

Urban communities suffer from a spiral of decay where poor conditions induce crime, and crime contributes to worsening conditions. Policies aim to aid blighted areas through investment; however, the literature overlooks how residents themselves invest and disinvest in their homes in relation to surrounding crime trends. This study uses panel data on crime and housing indicators from the Baltimore Police Department and the Baltimore Department of Housing and Community Development to model the relationship between crime and investment in residential properties in Baltimore City, Maryland, from 2015 to 2021. Contrary to the expected narrative, the study finds a significant increase in rehabilitation projects within higher crime areas, suggesting an unexpected pattern of investment. However, these areas also see fewer demolitions than anticipated and maintain a higher proportion of unoccupied homes, suggesting that investment and disinvestment can coexist. The policy implications for Baltimore's crime and vacancy problems are discussed.

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1. Introduction

Crime is one of the main determinants of neighborhood appeal. Typically, if the crime rate increases in a certain area, the value of nearby homes will drop (Ceccato & Wilhelmsson, 2011, 2012; Clark & Cosgrove, 1990; Ihlanfeldt and Mayock, 2010). But beyond shaping its surrounding environment, crime also responds to it. For example, resident reports of “collective efficacy” and a higher sense of community are associated with lower rates of crime and greater perceptions of safety (Sampson et al., 1997; Cancino, 2005; Cantora et al., 2016; Iyer et al., 2020). Conversely, tangible signs of disorder are associated with higher perceptions and rates of crime (Wilson & Kelling, 1982; Skogan, 1992; Sampson et al., 1997; Cantora et al., 2016).

Accordingly, areas which suffer from high rates of crime find themselves stuck in a spiral: crime begets disorder and decay; disorder and decay begets crime (Skogan, 1992). Much of the literature seeks to quantify the effects of this decay on the endogenous determinants of crime, such as the collective efficacy or disorder noted above. However, there is less focus on how crime relates to a specific avenue out of decay: investment in residential properties. Thus, I am interested in whether this type of investment in Baltimore— a city notorious for high rates of crime and a debilitating amount of vacant residential properties— is at all responsive to crime rates. The answer to this question could inform the evaluation of policies that aim to target both crime and housing instability, and whether they will deliver on their promises. For example, if investment is not responsive to crime, then policy makers should be careful to assume that crime prevention policies will have significant effects for moving communities out of blight, given that a lack of investment remains. Accordingly, my research question is as follows: How does crime relate to investment and disinvestment in residential properties? Based on the literature, I hypothesize that areas with lower crime will see higher rates of investment.

To model the relationship between crime and investment, I focus on three measures of investment: rehabilitation of properties, demolition of old properties, and occupancy of homes. I investigate the relationship these measures have with total crime, as well as the break-down of total crime into violent crime and property crime. I use panel data from 2015 to 2021, collected at the Community Statistical Area (CSA) level (Appendix A), for an OLS regression with two-way fixed effects that control for variation across years and CSAs. I also employ additional controls for race, ethnicity, age, education, and poverty status.

When looking at the relationship between crime and property rehabilitations, I find that as crime increases, rehabilitations also increase. This is a surprising finding which suggests that residents of areas with high crime may be interested in preventing their property from decaying. However, I also find that demolitions decrease on average in areas with higher crime, suggesting that dilapidated buildings still plague these areas, which may be contributing to more disorder and crime at the same time that residents are attempting to rehabilitate their properties. Finally, I find that in areas with higher crime, there are more unoccupied homes on average. This suggests that there may be opposite trends co-existing in Baltimore: even though more residents are investing in the rehabilitation of their properties in higher-crime areas, more residents may be abandoning these areas compared to lower-crime areas. The implications of these trends are discussed, and robustness checks affirm the significance of these findings.

This paper proceeds as follows: in the next section (2), I introduce the situation in Baltimore, a city characterized for its high crime rate and its preponderance of vacant and abandoned homes. In Section 3, I discuss the relevant literature on the spiral of decline associated with communities blighted with high crime and infrastructure issues. In Section 4, I introduce my data source for my model before moving to the discussion of the methods I use in

Section 5. I present my findings and corresponding discussions in Section 6. I conclude in Section 7 with the implications of these findings for the evaluation of present-day policies in Baltimore that seek to address crime and the vacancy crisis, as well as a discussion of some limitations of my model.

2. Background

2.1. Baltimore: A City of Vacancy and Crime

Baltimore stands out among East Coast cities for a rather grim reason— despite the population boom in other cities, Baltimore is shrinking (Miller & Little, 2022). A long history of redlining in Baltimore established a pattern of displacement for blacks and disinvestment in their communities (Pappoe, 2016). Meanwhile, white families were fleeing the cities for the alluring suburbs, leaving many houses vacant in their wake. This was a common trend for cities in the rust-belt— such as Detroit, Pittsburgh, Buffalo, and Cleveland— and Baltimore was no exception. Experiencing a loss in manufacturing jobs, and with the suburbs not far, these cities witnessed their populations dramatically drop, allowing vacancies to skyrocket. Specifically in Baltimore, the population declined by 34.6% between 1950 and 2012 (Whiteman, 2014).

Baltimore is still struggling with a unique set of challenges as it deals with its high rates of vacancy and housing insecurity. Mainly, it lacks the tools that other cities often have available, such as eminent domain, tax sale, and receivership. As observed by Miller and Little (2022), this deficiency perpetuates a vicious cycle wherein “The worst homes get worse, the same neighborhoods grow or decline, and the vacancy rates stay mostly stagnant.” Furthermore, these vacancies pose physical dangers to residents. In 2016, Baltimore resident Thomas Lemmon was

killed when the wall of an abandoned building collapsed, burying him and his Cadillac (Scott, 2020).

Beyond issues of housing insecurity, Baltimore struggles with endemic crime. In 2004, Baltimore's homicide rate was three times the rate in Los Angeles, and five times that of New York. The city was even the inspiration for an NBC police drama, "Homicide: Life on the Street" (Collins, 2007, p. 421). These conditions are better understood in relation to the history of discriminatory practices which kept people— blacks, specifically— from access to wealth building opportunities such as homeownership. In such circumstances, residents may need to turn to an informal economy in order to subsist (Pappoe, 2016, p. 131). It should not be a surprise, then, that a city that has experienced such disinvestment via segregationist policies would presently struggle with some of the most notable rates of crime in the nation (Dao, 2005).

At the heart of this problem lies the irony of urban inequality— that those populations most in need of revitalization and reinvestment find themselves continuously disenfranchised by policies and practices. As Badger (2015) observes about Baltimore in particular, it is "active decisions and government policies that have undermined the same people and sapped them of their ability to rebuild, that have again and again dismantled the same communities, each time making them socially, economically, and politically weaker." I argue, then, that studying how these communities respond to the circumstances these policies have forced upon them— ones of high crime and housing insecurity— may provide a new path to reduce this urban inequality.

2.2. Looking at Mechanisms to Induce Investment

Baltimore has long wrestled with ways to reduce its housing blight, most of which involve direct investment in infrastructure — redeveloping properties and demolishing

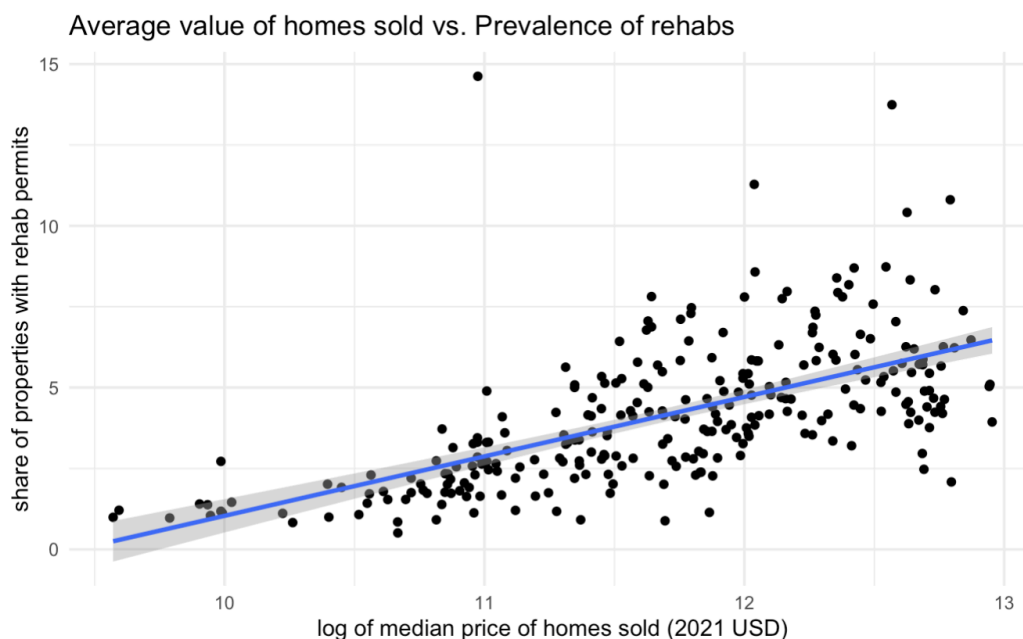
dilapidated ones, while physically marking the abandoned buildings that pose a threat to safety (Thompson et al., 2023). These policies, though, have been slow-moving and cannot fully address the community's needs, nor solve the dangerous conditions in a timely manner. This creates room for an inquiry— what else might influence housing investments, and provide a way out of this blight?

The urgency of this problem is exacerbated considering the positive relationship between physical disorder and crime (Newman, 1972; Newman & Franck, 1982; Wilson and Kelling, 1982). Though Baltimore has seen a slight decrease in crime since 2015 (Appendices F.1, F.2, F.3), it is far from a resolved issue in the city. And though the city has extensive programs and plans for addressing crime, as well as addressing vacancies, there have not been many impactful policies that acknowledge the interplay of these two issues.

Baltimore leaders are very aware of their housing and crime problems (even if they design policies which tackle them separately). To monitor these issues, the Baltimore Neighborhood Indicators Alliance (BNIA) publishes data on various “vital signs” to help the community measure progress towards meaningful outcomes at the community level (*Vital Signs 21*, n.d.). The data is aggregated at the Community Statistical Area level (Figure 1):

Among other categories, these vital signs provide valuable information related to housing that are proxies for blight, such as the percentage of properties which are vacant and abandoned, and the percentage of properties with tax liens that have been sold. There are also indicators which provide information on the level of investment in these residential properties. Below, I present the distribution of three of these indicators compared to the average value of homes sold in the corresponding neighborhood.

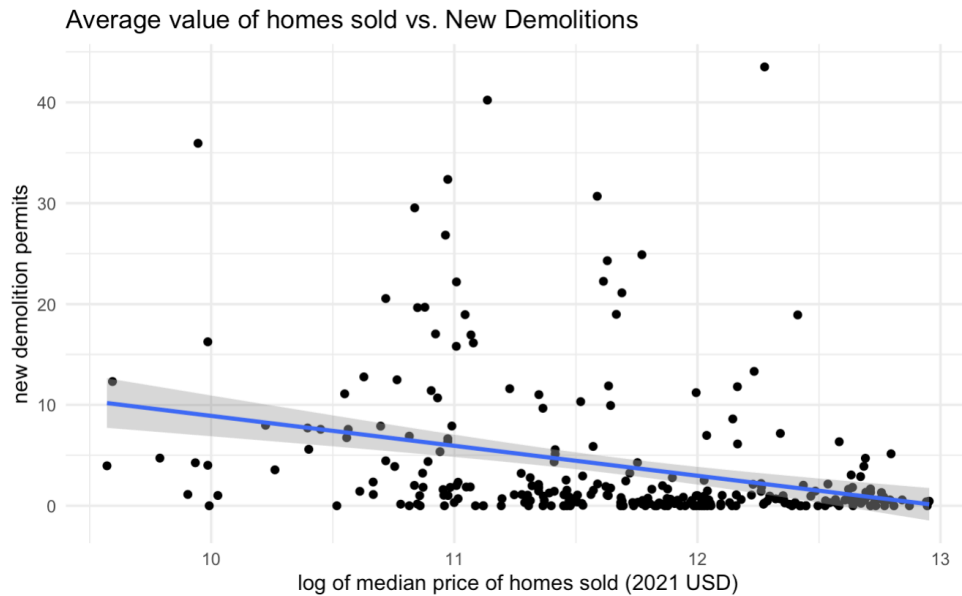
Figure 1. Rehabilitations are more prevalent in neighborhoods with more expensive homes.



Source: Author's calculations using pooled data from First American Real Estate Solutions (FARES) and DHCD data from 2015-2019. Calculations for adjusting prices to 2021 dollars can be found in Appendix C.3.

The share of rehabilitations captures how much residents are willing to invest in their existing assets. This is important because a person's home is typically their largest asset. If people are not continually investing in their homes, it may be a sign that they will be stuck in disrepair? This figure supports the existing literature on housing and urban decline, which asserts that as a population grows poorer, people will have less money to maintain their homes (Glaeser and Gyourko, 2001). This figure motivates my model because it begs the question — are rehabs only happening in areas which are already shielded from the negative effects of crime?

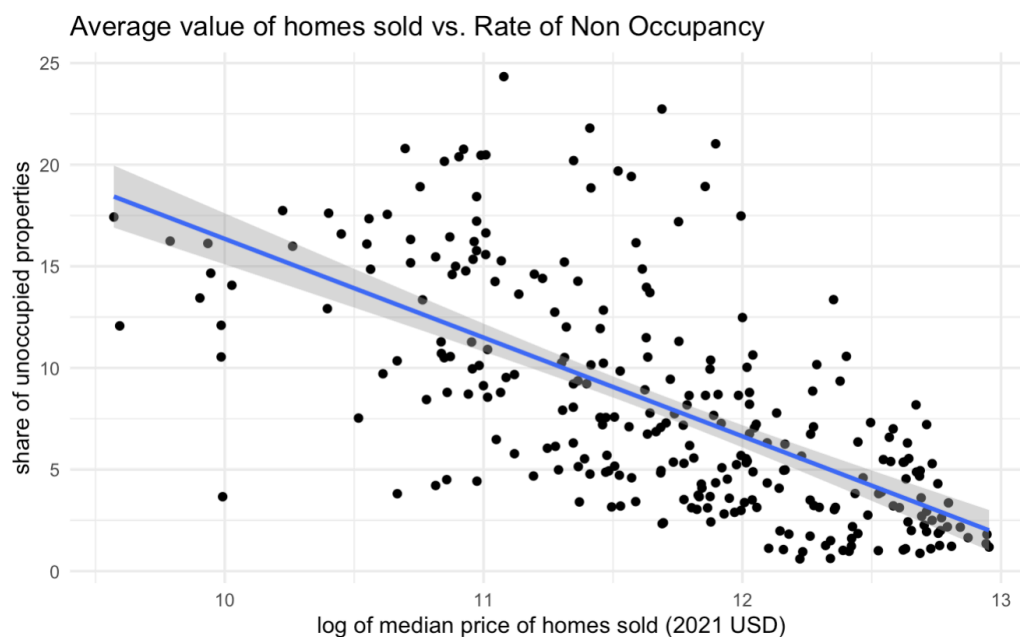
Figure 2. Neighborhoods with lower home values tend to have more demolitions.



Source: Author's calculations using pooled data from First American Real Estate Solutions (FARES) and DHCD data from 2015-2019. Calculations for adjusting prices to 2021 dollars can be found in Appendix C.3.

The share of demolitions captures the investment that the city is putting into these neighborhoods. For Baltimore, demolitions need to happen in order to get rid of the dangerous, unsound vacant structures. More demolitions then, is indicative of more investment.

Figure 3. Neighborhoods with lower home values suffer from higher rates of non-occupancy.



Source: Author's calculations using pooled data from First American Real Estate Solutions (FARES) and DHCD data from 2015-2019. Calculations for adjusting prices to 2021 dollars can be found in Appendix C.3.

The rate of unoccupied homes is an indicator collected by the USPS for homes which do not send or receive mail for at least 90 days.¹ If a home is not occupied, it may not be immediately dangerous or indicative of disinvestment in a neighborhood. However, it may not be long before people realize the property is abandoned, and the structure begins to deteriorate either because of a lack of maintenance or because people take advantage of the uninhabited property for their own activities. Thus, a higher rate of non occupancy is a proxy for a lack of investment and interest in a neighborhood. It is also interesting that many homes are unoccupied in low-value neighborhoods, because Baltimore also has a sizable homeless population. If many

¹ There is an important difference between “vacant and abandoned” homes and “unoccupied” homes. Whereas Baltimore uses “vacant and abandoned” to refer to buildings which are “uninhabitable” or no longer fit for someone to live in, “unoccupied” homes are structurally sound and habitable, but lack a resident.

of the homes that are unoccupied are on the more affordable side, Baltimore could explore programs to subsidize the payments on these properties to allow people to move in.

While these figures affirm the expectations that areas with lower home values have less investment — fewer rehabs, only a slightly higher amount of demolitions, and much higher rates of non occupancy— it also begs the question, do crime rates track this lack of investment? Or does the pattern of these housing indicators diverge from what we can expect given the value of homes? To get to the bottom of this, I will be looking precisely at how crime is related to indicators of investment.

The implications of my results will be important for understanding Baltimore's future and how it should prioritize investment in its neighborhoods not only to address housing vacancy and insecurity, but to address the blight that contributes to crime. As discussed, there is a feedback loop between disorder and crime. However, if Baltimore demonstrates unexpected associations between crime and investment, it could be an indicator that Baltimore is moving in a positive direction.

As will be detailed in my literature review, physical revitalization efforts may be especially effective for crime reduction, as residents often identify neighborhood-level disorder as a main contributor to crime (Cantora et al. 2016), and indeed, many studies look at the effect of property investment on crime levels (Citations!). This may also have a positive impact on housing development— as perceptions of safety improve and crime rates decline, the allure of these neighborhoods may increase. This could draw in more developers and residents, thereby injecting much-needed growth and community into these previously neglected areas.

3. Literature Review

There is a recognized problem of endogeneity when it comes to crime studies, especially in urban areas (Ihlanfeldt & Mayock, 2010; Mallach, 2016; Kuhlmann, 2023). The literature focuses on how crime is the result of blight, and how crime creates blight which then contributes to more crime, creating a cycle of disrepair. Before discussing the results of my study in the relationship crime has with potential avenues out of blight, I frame the role of crime in urban decline through a discussion of the existing literature. Most crime literature recognizes that crime is both a driver of urban decline, and an outcome of it. However, the literature overwhelmingly focuses on measuring results where crime is a dependent variable. For example, foundational work in theories of why crime occurs (e.g. Social Disorganization Theory and Broken Windows) explain crime as the outcome of localized structural and social characteristics. Despite the extensive focus on crime as an outcome, there are studies of crime as an explanatory variable, though most frequently as a determinant of home prices. I first summarize the literature where crime is posited as the outcome of several explanatory variables, both in increasing crime contexts and decreasing crime contexts. I then summarize the more brief literature on crime as the explanatory variable.

3.1. Crime as an Outcome

There are many academic contributions to the field of crime and criminal activity and many ideas about what drives crime. Certain themes do emerge, though, and an especially prevalent one is the role of the immediate environment. Within that immediate environment, two things have been extensively studied for their possible impact on crime: the tangible infrastructure, and the intangible sense of community. As will be explained in greater detail,

more appealing infrastructure and well-maintained neighborhoods may reduce crime (South et al., 2023). Furthermore, a greater sense of community and collective efficacy may also reduce crime (Sampson et al., 1997).

Crime as the result of infrastructure and the built environment (rehab, demos)

In 1972, Newman advanced the idea of “defensible space,” a theory that posits that the physical design of residential spaces has a significant impact on the incidence of crime, along with resident’s fear of future crime (Newman, 1972; Newman & Franck, 1982). This theory posits that when common spaces (e.g. community rooms in high-rise buildings) lack a clear owner, residents “cannot assert responsibility for their safety and maintenance, and these places are left vulnerable to crime and vandalism” (*Update from “Defensible Space” Pioneer Oscar Newman*, 1996). Research on this theory demonstrated that in low-income public housing projects, building height was among the top three most important predictors of robbery (Newman, 1973). Though these results have since been modified, the idea that architectural design and the surrounding environment may spur criminal behavior still informs research and even public opinion today. For example, a barrier to affordable housing development may be the widespread belief that the addition of affordable housing will both lower the property values of nearby homes and invite more crime (Hipp et al, 2022). However, in a study of Orange County, California, the placement of affordable housing units is correlated with *decreases* in various types of crime— including aggravated assault, robberies, and property crimes (ibid). Thus, while a lack of investment and attention to physical infrastructure may exacerbate crime, the expansion of housing access may indeed be a crime deterrent.

A reduction in crime as a result of investment in properties (Rehabilitating and Demolishing Properties)

Remedying physical conditions to reduce crime

In Philadelphia, a randomized experiment was recently conducted to empirically quantify the benefits of rehabilitating signs of physical disorder. The experiment tested whether the act of remediating abandoned properties can reduce crime. This study found that the properties with the highest level of remediation treatment had the biggest buffer against upward trends in gun violence (South et al, 2023). This lends credence to the idea that interventions on the physical level may be just as effective as interventions on the social level.

Looking at Demolitions and Rehabilitations (Similar Studies)

Existing literature on the relationship between investment and crime links together rehabilitations and demolitions in the conceptualizations of property investment (Smirniotis et al., 2022). This makes sense, considering that often, rehabilitation projects might rely on the demolition of old buildings before the new ones can take root. However, it would be a mistake to completely merge these two variables. I find that rehabilitations and demolitions have opposing relationships to crime rates (Figures 2 and 3), suggesting that the two activities do not completely overlap. Thus, my study will fill the gap in the literature created by the aggregation of rehabilitations and demolitions.

Demolitions

Han finds that “demolition of abandoned properties does not have any significant impact on nearby violent and property crime. This analysis shows that a change in nearby crime is attributable to differences in nearby socioeconomic and housing characteristics, rather than to the demolition of abandoned properties.”

Kulhmann (2020) looks at the impacts of demolition on investment and finds that “targeted demolitions of abandoned and distressed housing are associated with improvements in the physical condition of nearby houses. I find that properties located near demolitions are more likely than those near continuously vacant houses to have aspects of their external condition improve between the two inventories and less likely to experience their conditions deteriorate. I also find that properties located near houses owned, but not demolished, by the land bank are more likely to experience a deterioration in their external condition and less likely to experience improvements.”

- However, he measures the IMAPCT of investment, less so the outcome of investment
- Furthermore, he uses more subjective measures of the effects of demolition.

Spader et al. (2016) investigate the potential of demolitions to reduce crime in Cleveland, Ohio, and find that demolition activity in Cleveland decreased burglary and theft, but do not find measurable impacts of property rehabilitation investments—although the precision of these estimates are limited by the number of rehabilitation activities.

(Crime as the result of) Social Disorganization (occupancy?) and Broken Windows
(occupancy, rehabs, demos)

Shaw and McKay’s premier work on “social disorganization theory” posits that crime rates are based on the structural and environmental characteristics of a given area, factors that determine the level of “disorganization” in a neighborhood. Social disorganization can be defined as the inability of a community to realize the common values of its residents and

maintain effective social controls (Bursik, 1988), and is commonly found in neighborhoods with high unemployment or structural decay (“Social Disorganization Theory (Shaw & McKay)”). Furthermore, this theory argues that a lack of social cohesion inhibits the sense that the community can work toward achieving common goals. Because combating crime is often a goal for neighborhoods, socially disorganized communities are unfortunately ineffective in doing so. Meanwhile, crime is more successfully addressed in socially organized communities (“Shaw, Clifford R., and Henry D. McKay,” 2010, p. 829).

Shaw and McKay’s idea is echoed in one of the most infamous pieces of literature in the criminological sphere. In 1982, Wilson and Kelling released a piece in *The Atlantic*, detailing the framework of their “Broken Windows” theory of crime. They argue that, when a community appears to be in decline visually (via physical attributes such as literal broken windows, graffiti, and/or trash), that is an invitation for more crime to occur. Essentially, the theory holds that if a community signals that it has a tolerance of minor disorders, it is signaling to criminals that delinquent behavior will go unreported and uncontrolled— similar to Shaw and McKay’s idea that crime is attributable to a lack of shared values which would empower the reporting and monitoring of crime. This concept is captured in the “broken windows” metaphor by the notion that one unrepaired broken window sets a precedent that invites further vandalism, leading to a progressive breakdown of community standards and making the area more susceptible to crime.

Since this article was released, many other studies have sought to either build off its findings or critique them. One example of such a continuation of this theory comes from Skogan (1990), who explored a “spiral of decay” wherein low levels of social cohesion contribute to crime, and higher levels of crime also contribute to low levels of social cohesion and collective efficacy.

Efforts to increase a local sense of collective efficacy, as per the arguments of Skogan and Sampson, are important in augmenting social organization, which they argue is an important aspect of crime prevention. This argument was informally tested in a Baltimore neighborhood in 2013. Though the program's effect on crime is hard to isolate, the comprehensive set of initiatives in the program is argued to be the reason the neighborhood saw an "impressive" 500-day period without any homicides (Iyer et al., 2020, p. 24).

3.2. Crime as the Cause: Crime's Impact on Communities and Property Values

Despite the focus of crime literature on crime as an outcome variable, there is a branch of the literature which looks at crime as an explanatory variable, but specifically in relation to housing prices and neighborhood value. In accordance with the literature establishing a positive trend between disorder (physical and social) and crime, the reversal of this relationship yields similar results. That is, the prevalence of crime creates a sense of decline, which decreases demand for properties and homes in the neighborhood (Buonanno et al., 2012), which is then demonstrated by lower property values in higher crime neighborhoods (Thaler, 1978; Ceccato & Wilhelmsson, 2011, 2012; Clark & Cosgrove, 1990; Ihlanfeldt and Mayock, 2010). Furthermore, crime and the less-tangible fear of crime lead to flight from the city into the suburbs (Massey & Denton, 1993; Tita et al., 2006). Simply put, the literature on crime asserts that as crime increases, neighborhood values will decline.

3.3. Gap in literature

Scholars consistently look at the explanatory power that rehabilitations may have for things such as crime (Han) or more residential investment (Kuhlmann), but there is less focus on

how this crime itself might explain these investments. Because crime and investment are endogenous (right word?), it is worth looking at the reverse direction of the relationship in order to understand the magnitude of the effect of one on the other. In doing so, policy makers can have a better understanding of whether they should funnel more efforts into property rehabilitations or in crime prevention policies if they want to tackle the problems of urban decline.

Raleigh & Galster (2016) study crime as an outcome of disinvestment and abandonment, contributing to the literature which is focused on crime as an outcome, and less so as an explanation of disinvestment.

Despite the preponderance of crime literature, there is a gap in relation to crime's effect on more dynamic indicators of neighborhood conditions that signal a change in the area— such as investment. My study fills this gap by looking directly at how crime relates to two key measures of physical investment— rehabilitations and demolitions— as well as looking at occupancy as a measure of social investment.

To my knowledge, only one other study has looked at crime as an explanatory variable for property investment (Lacoe et al., 2018). They find an inverse relationship between crime and investment, specifically in areas experiencing reductions in crime. Their measure of investment was through the issuance of new building permits in aggregate. My study builds on this by examining permits which are specifically for rehabilitation of existing properties, and those which are for demolition of existing properties. The breakdown of these permits allows me to estimate whether the investment being made is ... (if rehabs happen in more expensive areas, and demos happen in poorer areas, then more demos would mean more investment for poor communities?

4. Data

To study how crime relates to residential investment, I use Baltimore's Department of Housing and Community Development (DHCD) data, Baltimore Police Department (BPD) data, and Census data from the American Community Survey (ACS). These data are downloaded from the Baltimore Neighborhood Indicators Alliance (BNIA) Open Data Portal. BNIA collects data from reputable sources and publishes "Vital Signs" on this portal to help the community measure progress towards meaningful outcomes at the community level (*Vital Signs 21*, n.d.). Data is available for years 2015-2021 for over 150 indicators. Based on my review of the literature, I focus my study on three key variables which capture investment through their role in maintaining physical and social order: 1) the prevalence of rehabilitation permits, 2) the prevalence of demolition permits, and 3) the prevalence of unoccupied homes. These variables serve as the dynamic proxies for investment that will help answer my question of how crime relates to investment and disinvestment in residential properties. Table 1 presents the name and category of each variable I use, the source description of how this variable is measured, and the available years I used in my model.

The data use Community Statistical Areas (CSAs) as the unit of observation rather than census tracts. CSAs are designed to capture demographic homogeneity while reflecting resident perceptions of community boundaries (Appendix A). Thus for my study, this unit of analysis should be sufficiently granular to capture the relationship between crime and investment in a particular CSA without systematically masking minority trends.

My first outcome variable of interest is the share of residential properties within a CSA that have a permit for a rehabilitation project exceeding \$5,000 in total cost. The threshold of \$5,000 is used by the DHCD to differentiate minor renovations from more significant projects.

Thus, the count of rehabilitations captured in this variable represent more substantial projects than a new coat of paint. Given that the median household income in Baltimore was \$52,318 in 2021, a \$5,000 rehabilitation project would pull about 10% of the average resident's annual income. Hence, this variable is a strong proxy for an owner's investment in their property and physical order.

Data on demolitions also comes from Baltimore DHCD. These data show the number of permits issued for the demolition of residential buildings per 1,000 existing residential properties. The city uses demolitions to address vacant properties, specifically those which are so unstable that they pose a safety risk to residents walking nearby (*Baltimore's Collection Of Vacant Houses*, 2022). Demolitions are an important tool for the city to address vacancies, and thus represent investment in improving and maintaining physical order.

For capturing my third variable of interest, social investment, I use data on unoccupied homes collected by USPS and Baltimore DHCD. Homes are considered unoccupied when the postal service finds that there has been no mail collection (sent or received) at the property for at least 90 days. Though there are some limitations for this data in measuring non occupancy—for example, units may be counted as unoccupied if the residents receive their mail at a separate PO Box (Molloy, 2016)—this data has been used to represent vacant or unoccupied homes in a number of studies (Molloy, 2016; Immergluck, 2016; Wang & Immergluck, 2019). Furthermore, Silverman et al. (2013) list many arguments for the advantages of using USPS data to measure non-occupancy, rather than census data. Among them are the benefits of having current data based on full counts of all addresses in an area, and the benefits of having data from a source that subdivides properties by residential, business, and other types.

Furthermore, even though electronic forms of communication are increasingly used, snail mail is still an essential service that remains critical for the delivery of official documents such as jury summons and mail ballots. If there is a large portion of citizens who are not receiving mail, even if they are still occupying their home, it may be a sign that they are not engaging with their community. Thus, higher rates of non-occupancy may capture lower levels of social investment.

The crime statistics I use as my explanatory variable are reported by the Baltimore Police Department. Baltimore classifies most major crimes— homicide, rape, aggravated assault, robbery, burglary, larceny, and auto theft— as “Part 1” crime. BPD also reports whether the crime is identified as violent (homicide, rape, aggravated assault, and robbery) or property-related (burglary, larceny, and auto theft). Hence, shares of violent and property crime should add up to total Part 1 crime. Data is reported as the number of incidents per 1,000 residents. To normalize the distribution of crime across CSAs, I modify this rate to reflect instead the number of incidents per the population within each CSA (Appendix C.3-5). I touch on some limitations of this crime data in Section 7.2.

Control variables for race, ethnicity, age, education, employment, and poverty status are obtained from the American Community Survey, an ongoing survey conducted by the US Census Bureau to capture moving averages of demographic and housing indicators. The survey is administered every year, and is the best representation of up-to-date demographic and housing conditions at the CSA level.

Table 1. Description of Variables in Model

Variable Name	Variable Type	Description	Source and Years
HOUSING			

Percentage of Properties with Rehabilitation Permits Exceeding \$5,000	Outcome	The percent of residential properties that have applied for and received a permit to renovate the interior and/or exterior of a property where the cost of renovation will exceed \$5,000. The threshold of \$5,000 is used to differentiate a minor and more significant renovation project.	Source: Baltimore Department of Housing and Community Development Years Used: 2015, 2016, 2017, 2018, 2019, 2020, 2021
Number of Demolition Permits per 1,000 Residential Properties	Outcome	The number of permits issued for the demolition of residential buildings per 1,000 existing residential properties. The permits are analyzed by date of issue and not date of actual demolition.	Source: Baltimore Department of Housing and Community Development Years Used: 2015, 2016, 2017, 2018, 2019, 2020, 2021
Percent of Residential Properties that are Unoccupied	Outcome	The percentage of residential addresses for which the United States Postal Service has identified as being unoccupied (no mail collection) for a period of at least 90 days or longer.	Source: U.S. Postal Service, U.S. Department of Housing and Urban Development Years Used: 2015, 2016, 2017, 2018, 2019, 2020
Median Price of Homes Sold	Descriptive	The median home sales price is the middle value of the prices for which homes are sold (both market and private transactions) within a calendar year. The median value is used as opposed to the average so that both extremely high and extremely low prices do not distort the prices for which homes are sold. This measure does not take into account the assessed value of a property.	Source: First American Real Estate Solutions (FARES) Years Used: 2015, 2016, 2017, 2018, 2019
CRIME			
Part 1 Crime Rate	Explanatory	The Part 1 crime rate captures incidents of homicide, rape, aggravated assault, robbery, burglary, larceny, and auto theft that are reported to the Police Department, per number of residents in the CSA (calculation in Appendix C.3).	Source: Baltimore Police Department Years Used: 2015, 2016, 2017, 2018, 2019, 2020, 2021
Violent Crime Rate	Explanatory	Part 1 crimes identified as being violent (homicide, rape, aggravated assault, and robbery), per number of residents in the CSA (calculation in Appendix C.4).	Source: Baltimore Police Department Years Used: 2015, 2016, 2017, 2018, 2019, 2020, 2021

Property Crime Rate	Explanatory	Part 1 crimes identified as being property-based (burglary and auto theft), per number of residents in the CSA (calculation in Appendix C.5).	Source: Baltimore City Police Department Years Used: 2015, 2016, 2017, 2018, 2019, 2020, 2021
RACE			
Percent Black	Control	The total number of persons that identify themselves as being racially Black or African American (and ethnically non-Hispanic) out of the total number of persons living in an area. "Black or African American" refers to a person having origins in any of the Black racial groups of Africa. It includes people who indicated their race as "Black".	Source: U.S. Census Bureau, American Community Survey Years Used: 2011-2015, 2012-2016, 2013-2017, 2014-2018, 2015-2019, 2020, 2017-2021
Percent White	Control	The percentage of persons, out of the total number of persons living in an area, self-identifying as racially White (and ethnically non-Hispanic). "White" refers to a person having origins in any of the original peoples of Europe, the Middle East, or North Africa. This indicator includes people who identified their race(s) as "White".	Source: U.S. Census Bureau, American Community Survey Years Used: 2011-2015, 2012-2016, 2013-2017, 2014-2018, 2015-2019, 2020, 2017-2021
Percent Hispanic	Control	The percentage of persons, out of the total number of persons living in an area, self-identifying their ethnicity as Hispanic or Latino. Hispanic origin can be viewed as the heritage, nationality group, lineage, or country of birth of the person or the person's parents or ancestors before they arrived in the United States. People who identify their origin as Hispanic, Latino, or Spanish may be of any race.	Source: U.S. Census Bureau, American Community Survey Years Used: 2011-2015, 2012-2016, 2013-2017, 2014-2018, 2015-2019, 2020, 2017-2021
Percent Asian	Control	The percentage of persons, out of the total number of persons living in an area, self-identifying as Asian (and non-Hispanic). "Asian" refers to a person having origins in any of the original peoples of the Far East, Southeast Asia, or the Indian subcontinent, including, for example, Cambodia, China, India, Japan, Korea, Malaysia, Pakistan, the Philippine Islands, Thailand, and Vietnam.	Source: U.S. Census Bureau, American Community Survey Years Used: 2011-2015, 2012-2016, 2013-2017, 2014-2018, 2015-2019, 2020, 2017-2021

Percent two or more races	Control	The percentage of persons, out of the total number of persons living in an area, who self-identify as being of two or more races (non-Hispanic).	Source: U.S. Census Bureau, American Community Survey Years Used: 2011-2015, 2012-2016, 2013-2017, 2014-2018, 2015-2019, 2020, 2017-2021
Percent all other races	Control	The percentage of persons, out of the total number of persons living in an area, who self-identify as either American Indian, Alaskan Native, Native Hawaiian or Other Pacific Islander, or some other race (non-Hispanic).	Source: U.S. Census Bureau, American Community Survey Years Used: 2011-2015, 2012-2016, 2013-2017, 2014-2018, 2015-2019, 2020, 2017-2021
AGE			
Percent of Population Under 5 Years old	Control	The percent of persons, out of all persons living in an area, aged 5 years and under.	Source: U.S. Census Bureau, American Community Survey Years Used: 2011-2015, 2012-2016, 2013-2017, 2014-2018, 2015-2019, 2016-2020, 2017-2021
Percent 5-17	Control	The percent of persons, out of all persons living in an area, aged 5 to 17 years.	Source: U.S. Census Bureau, American Community Survey Years Used: 2011-2015, 2012-2016, 2013-2017, 2014-2018, 2015-2019, 2016-2020, 2017-2021
Percent 18-24	Control	The percent of persons, out of all persons living in an area, aged 18 to 24 years.	Source: U.S. Census Bureau, American Community Survey Years Used: 2011-2015, 2012-2016, 2013-2017, 2014-2018, 2015-2019, 2016-2020, 2017-2021
Percent 25-64	Control	The percent of persons, out of all persons living in an area, aged 25 to 64 years.	Source: U.S. Census Bureau, American Community Survey Years Used: 2011-2015, 2012-2016, 2013-2017, 2014-2018, 2015-2019, 2016-2020, 2017-2021
Percent over 65	Control	The percent of persons, out of all persons living in an area, 65 years and above.	Source: U.S. Census Bureau, American Community Survey Years Used: 2011-2015, 2012-2016, 2013-2017,

			2014-2018, 2015-2019, 2016-2020, 2017-2021
EDUCATION			
Percent less than high school diploma	Control	The percentage of persons that have not completed, graduated, or received a high school diploma or GED. This is a standard indicator used to measure the portion of the population with less than a basic level of skills needed for the workplace. Persons under the age of 25 are not included in this analysis since many of these persons are still attending various levels of schooling.	Source: U.S. Census Bureau, American Community Survey Years Used: 2011-2015, 2012-2016, 2013-2017, 2014-2018, 2015-2019, 2016-2020, 2017-2021
Percent Population (25 Years and over) With High School Diploma and Some College or Associate's Degree	Control	The percentage of persons that have completed, graduated, or received a high school diploma or GED and also have taken some college courses or completed their Associate's degree. This is a standard indicator used to measure the portion of the population with a basic level of skills needed for the workplace. Persons under the age of 25 are not included in this analysis since many of these persons are still attending various levels of schooling.	Source: U.S. Census Bureau, American Community Survey Years Used: 2011-2015, 2012-2016, 2013-2017, 2014-2018, 2015-2019, 2016-2020, 2017-2021
Percent Population (25 Years and over) with a Bachelor's Degree or Above	Control	The percentage of persons that have completed, graduated, or received a Bachelor's or an advanced degree. This is an indicator used to measure the portion of the population having an advanced level of skills needed for the workplace. Persons under the age of 25 are not included in this analysis since many of these persons are still attending various levels of schooling.	Source: U.S. Census Bureau, American Community Survey Years Used: 2011-2015, 2012-2016, 2013-2017, 2014-2018, 2015-2019, 2016-2020, 2017-2021
EMPLOYMENT			
Percent Population 16-64 Unemployed and Looking for Work	Control	The percent of persons between the ages of 16 and 64 not working out of all persons, not just those in the labor force (persons who may be looking for work). These persons are seeking work that pays a formal income.	Source: U.S. Census Bureau, American Community Survey Years Used: 2011-2015, 2012-2016, 2013-2017, 2014-2018, 2015-2019, 2016-2020, 2017-2021
Percent Population 16-64 Employed	Control	The percent of persons between the ages of 16 and 64 formally employed or self-employed and earning a formal income.	Source: U.S. Census Bureau, American Community Survey

		It is used to understand how many persons are working out of the entire population, not just those in the labor force (persons who may be looking for work or working).	Years Used: 2011-2015, 2012-2016, 2013-2017, 2014-2018, 2015-2019, 2016-2020, 2017-2021
INCOME			
Median Household Income	Control	Median household income is the middle value of the incomes earned in the prior year by households in an area. Income is adjusted to 2021 dollars (Appendix C.2). The median value is used as opposed to the average so that both extremely high and extremely low prices do not distort the total amount of income earned by households in an area.	Source: U.S. Census Bureau, American Community Survey Years Used: 2011-2015, 2012-2016, 2013-2017, 2014-2018, 2015-2019, 2016-2020, 2017-2021
POVERTY			
Percent of Family Households Living Below the Poverty Line	Control	Percent of family households living below the poverty line measures the percentage of households, out of all households in an area, whose income fell below the poverty threshold. Federal and state governments use such estimates to allocate funds to local communities. Local communities use these estimates to identify the number of individuals or families eligible for various programs.	Source: U.S. Census Bureau, American Community Survey Years Used: 2011-2015, 2012-2016, 2013-2017, 2014-2018, 2015-2019, 2016-2020, 2017-2021

5. Methodology

To examine how crime is related to investment and disinvestment in residential properties, I run three OLS models with two-way fixed effects— one with rehabilitations as the dependent variable, one with demolitions as the dependent variable, and one with non-occupancy as the dependent variable. The two-way fixed effects control for variation across time and across CSAs. I also employ additional controls for race, ethnicity, age, education, employment, and poverty status. By including these controls, I am improving the explanatory power of crime in my model.

I have a panel data set with 7 years of data (2015-2021) for 55 Community Statistical Areas (CSAs), giving me a balanced panel of 385 observations. One exception is the data on housing non-occupancy; there is no data available for 2021. To account for this, the model for this variable only uses the balanced panel of 330 observations, from 2015-2020.

I run each model three times, once for each breakdown of crime: total Part 1 crime, Part 1 crime that is violent, and Part 1 crime that is property-related. An important consideration when working with crime data as an explanatory variable is the possibility of lagged responses. For reasons of both feasibility and delayed reactions, residents may be slow to invest or disinvest in properties following a crime incident. Laco et al. (2018) consider this when they study the impact of crime on building investment in Chicago and Los Angeles, and find that a one-year lag for crime is appropriate. Accordingly, I use a one-year lag structure by matching crime from year $t-1$ with the observation of the dependent variable in time t . By doing so, I allow the model to more precisely capture resident investment as related to neighborhood crime incidents.

5.1. Percent of Properties with Rehabilitation Permits

I first consider the relationship that total crime has with the percentage of properties with rehabilitation permits. I then break up this analysis to find the relationship between rehabilitations and violent crime, as well as rehabilitations and property crime.

$$\%rehab_{CSA,t} = \beta_1 part\ 1\ crime_{CSA,t} + \sum_{i=2...K} \beta_i X_{CSA,t} + \varphi_{CSA} + \delta_t + \varepsilon_{CSA,t} \quad (\text{eq. 1})$$

$$\%rehab_{CSA,t} = \beta_1 violent\ crime_{CSA,t} + \sum_{i=2...K} \beta_i X_{CSA,t} + \varphi_{CSA} + \delta_t + \varepsilon_{CSA,t} \quad (\text{eq. 2})$$

$$\%rehab_{CSA,t} = \beta_1 property\ crime_{CSA,t} + \sum_{i=2...K} \beta_i X_{CSA,t} + \varphi_{CSA} + \delta_t + \varepsilon_{CSA,t} \quad (\text{eq. 3})$$

where $X_{CSA,t}$ is a vector of control variables representing shares of the population by: age, race, ethnicity, education attainment, employed status, and living below the poverty line, φ_{CSA} accounts for the effects on the percentage of properties with rehabilitation permits that are constant within community statistical areas over the timeframe of the study and that have an additive impact (such as houses located near a body of water, park, or major infrastructure). The year fixed effects, δ_t , account for macroeconomic factors (such as a recession or inflation) that affect the percentage of properties with rehabilitation permits similarly across all community statistical areas but differ by year.

5.2. Percent of Properties with Demolition Permits

Next, I model the relationship between the three types of crime and the rate of demolition permits.

$$demolitions_{CSA,t} = \beta_1 part\ 1\ crime_{CSA,t} + \sum_{i=2...k}^K \beta_i X_{CSA,t} + \varphi_{CSA} + \delta_t + \varepsilon_{CSA,t} \quad (eq. 4)$$

$$demolitions_{CSA,t} = \beta_1 violent\ crime_{CSA,t} + \sum_{i=2...k}^K \beta_i X_{CSA,t} + \varphi_{CSA} + \delta_t + \varepsilon_{CSA,t} \quad (eq. 5)$$

$$demolitions_{CSA,t} = \beta_1 property\ crime_{CSA,t} + \sum_{i=2...k}^K \beta_i X_{CSA,t} + \varphi_{CSA} + \delta_t + \varepsilon_{CSA,t} \quad (eq. 6)$$

where $X_{CSA,t}$ is a vector of control variables representing shares of the population by: age, race, ethnicity, education attainment, employed status, and living below the poverty line, φ_{CSA} accounts for the effects on the **rate** of demolition permits that are constant within community statistical areas over the timeframe of the study and that have an additive impact (such as houses located near a body of water, park, or major infrastructure). The year fixed

effects, δ_t , account for macroeconomic factors (such as a recession or inflation) that affect the rate of demolition permits similarly across all community statistical areas but differ by year.

5.3. Prevalence of Unoccupied Residential Properties

I then look at the relationship between the three types of crime and the prevalence of unoccupied properties.

$$\% \text{ unoccupied}_{CSA,t} = \beta_1 \text{ part 1 crime}_{CSA,t} + \sum_{i=2 \dots k}^K \beta_i X_{CSA,t} + \varphi_{CSA} + \delta_t + \varepsilon_{CSA,t} \quad (\text{eq. 7})$$

$$\% \text{ unoccupied}_{CSA,t} = \beta_1 \text{ violent crime}_{CSA,t} + \sum_{i=2 \dots k}^K \beta_i X_{CSA,t} + \varphi_{CSA} + \delta_t + \varepsilon_{CSA,t} \quad (\text{eq. 8})$$

$$\% \text{ unoccupied}_{CSA,t} = \beta_1 \text{ property crime}_{CSA,t} + \sum_{i=2 \dots k}^K \beta_i X_{CSA,t} + \varphi_{CSA} + \delta_t + \varepsilon_{CSA,t} \quad (\text{eq. 9})$$

where $X_{CSA,t}$ is a vector of control variables representing shares of the population by: age, race, ethnicity, education attainment, employed status, and living below the poverty line, φ_{CSA} accounts for the effects on the prevalence of unoccupied properties that are constant within community statistical areas over the timeframe of the study and that have an additive impact (such as houses located near a body of water, park, or major infrastructure). The year fixed effects, δ_t , account for macroeconomic factors (such as a recession or inflation) that affect the prevalence of unoccupied properties similarly across all community statistical areas but differ by year.

5.4. Assumptions of the model

There are two key assumptions of these models. The first is that the prevalence of each of the dependent variables (rehabilitation permits, demolition permits, and unoccupied houses) is affected linearly by the independent variables in the model. The second is that the unobservable factors affecting the prevalence of the dependent variables can be decomposed into components

that vary over time but not across community statistical areas (year fixed effects), components that vary across community statistical areas but do not vary across time (community statistical area fixed effects), and an idiosyncratic component that varies over time and across community statistical areas and is not correlated with the independent variables in the model.

Though I control for a wide set of variables, the design of my model does not allow for causal interpretation of the relationship between crime and investment. However, it is important to note that even though my model is not causal, results are not necessarily symmetric. Specifically, it is not valid to assert that the relationship of crime influencing investment can simply be reversed to imply that investment similarly influences crime. The mechanisms driving each direction of influence may differ significantly. This distinction is crucial for my discussion of the results, as I will explore the potential implications of the observed relationship between crime and investment on the hypothesized, yet unobserved, effects of investment on crime and disorder in Baltimore.

5.5. Choropleth Maps

I created maps of Baltimore city's CSAs using data from 2015-2021 (unless otherwise indicated). I used the quantile classification method to create equal proportions of data in each of the 5 categories for the first year of data displayed on the map. Then, to indicate how the data change over time, I impose the quantiles on the remaining years of data for that variable (Brewer & Pickle, 2002). When examining the maps, one can clearly see variation between CSAs and the trends over time.

6. Results and Discussion

The results proceed as follows: first, I discuss some descriptive statistics about the demographic makeup across the different CSAs in Baltimore, pertaining specifically to the ones that are significant controls in my model. I then move to the analysis of my models, beginning with the relationship between lagged crime and property rehabilitations. I follow that with the discussion of the relationship between lagged crime and the rate of unoccupied housing. Finally, I discuss the relationship between lagged crime and the rate of new demolitions.

In my discussion of the results, I focus on the immediate relationship crime has with the different proxies for investment as demonstrated in my model, and I discuss the implications of these relationships for the future outlook of Baltimore in relation to crime and housing security. My findings are mostly aligned with the patterns discussed in existing crime literature, which at a high level demonstrate that physical and social disorder go hand-in-hand with crime. However, my results contain a notable exception—I find that in areas with higher crime, there may be a *reduction* in physical disorder via a higher prevalence of rehabilitations. I discuss what this surprising result means in the context of Baltimore City and the implications for Baltimore’s policy plans for crime and housing market revitalization.

6.1. Descriptive Statistics

Because I am working with panel data, the best way to display the demographics of Baltimore is with maps. [Figure 5](#) displays the share of black residents in each CSA during my study period (2015-2021). The shades correspond to quantile break ups of the data, based on 2015 distributions. [Figure 6](#) displays the share of white residents in the same manner. Note that the areas with a majority of white residents are the same areas where blacks are significantly in the minority. Note also that the distribution of black residents is right-skewed—the second

quartile already captures majority-black neighborhoods. On the other hand, the distribution for whites is left-skewed, with only the top quintile capturing majority white neighborhoods. Figure 7 demonstrates the share of all other races. There is a noticeable change in the distribution and placement of this population compared to 2015 levels, but this is likely due to this group being a very small share of Baltimore's total population — never more than 6% in any one CSA. Therefore, the distribution of this group will be a bit more sensitive to noise.

Figure 8 displays the share of residents living below the poverty line. This distribution represents the number of families whose annual income falls below the threshold for poverty set by the federal government. Though the official poverty thresholds do not vary geographically, they are updated for inflation. Poverty measures are not intended to be a complete measure of what a family needs for subsistence; rather, they are intended for statistical analysis (*How the Census Bureau Measures Poverty*, n.d.). Therefore, this map demonstrates the distribution of income across Baltimore. Note that areas with much higher shares of families in poverty overlap with areas of higher populations of black residents. By the same coin, areas with low shares of families in poverty co-occur with higher proportions of white residents.

Figure 9 displays the distribution of crime in Baltimore, from 2015 to 2021. CSAs with higher rates of crime per capita indeed align with CSAs that have higher shares of residents in poverty.

6.2. Percent of Properties with Rehabilitation Permits

Table 2 shows that crime has a significant positive relationship with the prevalence of rehabilitation in the following year. If total crime increases by 10%, the rate of rehabilitation permits increases by 0.5% on average. Looking specifically at violent crime, a 10% increase is associated with a 1.4% increase in rehabilitations on average. Finally, a 10% increase in property

crime is associated with a 0.6% increase in rehabilitations on average. All results are statistically significant.

Choropleth maps of the change in rehabilitation rates across years and CSAs show a noticeable increase in the prevalence of rehabilitations from 2015 to 2021 (Appendix F.4). Notably, there does not seem to be an obvious spatial pattern in the increase in rehabilitations, suggesting they increased indiscriminately throughout the city.

The coefficient on the control variable for the share of black residents in the population is significant at the 1% level for all models (Appendix D.1). Interestingly, the sign of the coefficient becomes negative for this control, suggesting that among areas with more crime, those with higher shares of black residents see fewer rehabilitations.

Table 2. Relationship between lagged crime and the prevalence of rehabilitation permits in a CSA

Crime and Rehabilitation Permits			
	<i>Dependent variable:</i>		
	Rate of Rehabilitation Permits		
	(1)	(2)	(3)
Lag of Total Crime	0.050** (0.023)		
Lag of Violent Crime		0.144** (0.069)	
Lag of Property Crime			0.057* (0.032)
Demographic Controls	Yes	Yes	Yes
Year Fixed Effects	Yes	Yes	Yes
CSA Fixed Effects	Yes	Yes	Yes
# of CSAs	55	55	55
Observations	384	384	384
R ²	0.084	0.082	0.079
Adjusted R ²	-0.136	-0.137	-0.142
<i>Note:</i>	* p<0.1; ** p<0.05; *** p<0.01		

Notes: Standard errors are in parentheses. Levels of statistical significance are indicated at * p<0.1, ** p<0.05, *** p<0.01. Standard errors are in parentheses. 384 observations are used from the full panel of 385 observations due to the one-year lag on crime. Results rendered with stargazer (Hlavac, 2022). Source: Author's calculations using Baltimore Department of Housing and Community Development, Baltimore Police Department, and US Census American Community Survey data, 2015-2021

The results indicate that areas with higher crime tend to have more rehabilitation, suggesting that these higher-crime areas may be seeing more investment than lower-crime areas. This is a surprising finding, given that the literature establishes a relationship where crime is inversely related to physical order (Wilson and Kelling, 1982). This relationship would suggest that crime then facilitates a lack of investment, as found by Lacoé et al. (2018) in their studies of Chicago and Los Angeles. However, my findings indicate an opposite relationship— that investment happens even in areas with more prevalent crime.

This is a promising result for the future of these communities. Though my study does not identify the mechanisms by which this investment is being promoted, it nonetheless shows that high-crime areas may soon be benefitting from physical improvements as a result of the higher rates of rehabilitation permits.

On the other hand, though, my model cannot eliminate the possibility that the trend has appeared this way long before 2015, which would suggest that even though property rehabilitations are occurring at higher rates, they have not been impactful in moving these neighborhoods out of blight if the same high-crime communities are still experiencing high rates of crime, despite the higher rates of rehabilitations. However, given that Figure 1 shows rehabilitations occurring more frequently in areas with higher home prices, it may be the case that these rehabilitations are sufficient enough to overcome the price burden of high crime. If houses are selling for more, then, this may signal that the neighborhood is no longer tolerant of crime (CITE), and thus a slow progression out of the “spiral of decay” may be in motion. In the absence of a model which investigates the change in the relationship across years, however, it is hard to come to this conclusion for certain. Regardless, the significant positive relationship between crime and rehabilitation provides compelling evidence that the relationship between crime and physical order, as captured via rehabilitations, does not always align with the theories from literature.

One potential explanation for this surprising result would be the presence of an exogenous policy which incentivized rehabilitation in more blighted neighborhoods. However, to the best of my knowledge, such a policy did not exist in Baltimore during the period of the study. Another potential hypothesis is that people are purchasing cheap properties in blighted areas to renovate them and sell them for profit. However, if this were the case, it seems unlikely that the

association between prices of homes sold and rehabilitation rate would be upward-sloping as is shown in Figure 1. Rather, the relationship may be flat— as more rehabs happen in lower-value neighborhoods, the prices of these improved homes may drive up the median price of homes sold, shifting up the left-hand-side of the line of best fit between home prices and rehabilitation rates.

The significant negative coefficient on the control variable for the share of black residents is also noteworthy (Appendix D.1). This suggests that areas with more black residents experience less investment. The maps showing the rehabilitation rates and the share of black residents (Appendices F.4 and F.7) provide some visual validation of this result. Particularly, the CSAs in the central-western part of the city, which tend to be majority black, do not reach the top quintile for rehabilitation permits until 2020, indicating a period of disproportionate underinvestment.

With Baltimore's history of redlining, perhaps this result is not surprising. However, with such a large share of black residents— 57% of the population in 2020— it suggests that a significant portion of the city's population may be living in areas that are not receiving equitable investment via housing and infrastructure rehabilitation. This disparity contributes to the spiral of decline that rehabilitation would be poised to address, especially given that my model shows higher-crime areas receiving more rehabilitation, which is associated with higher home values (Figure 1). Encouraging rehabilitations in these areas may address the historical inequities and allow black homeowners to build their wealth— a practice from which they were systematically denied in the 20th century.

6.3. Percent of Properties with Demolition Permits

Table 3 shows that as crime increases, demolitions in the following year decrease on average. Specifically, as total crime increases by 10%, demolitions decrease by 2.1% on average. For violent crime specifically, a 10% increase coincides with a 6.8% decrease in demolitions. For property crime, a 10% increase is associated with a 2.5% decrease in demolitions. All results are statistically significant at the 5% level.

Table 3. Relationship between lagged crime and the prevalence of new demolition permits in a CSA

Crime and Demolitions			
	<i>Dependent variable:</i>		
	Rate of Demolition Permits		
	(1)	(2)	(3)
Lag of Total Crime	-0.205*** (0.077)		
Lag of Violent Crime		-0.676*** (0.232)	
Lag of Property Crime			-0.245** (0.109)
Demographic Controls	Yes	Yes	Yes
Year Fixed Effects	Yes	Yes	Yes
CSA Fixed Effects	Yes	Yes	Yes
# of CSAs	55	55	55
Observations	384	384	384
R ²	0.043	0.047	0.037
Adjusted R ²	-0.187	-0.181	-0.194
<i>Note:</i>	* p<0.1; ** p<0.05; *** p<0.01		

Notes: Standard errors are in parentheses. Levels of statistical significance are indicated at * p<0.1, ** p<0.05, *** p<0.01. Standard errors are in parentheses. 384 observations are used from the full panel of 385 observations due to the one-year lag on crime. Results rendered with stargazer (Hlavac, 2022). Source: Author's calculations using Baltimore Department of Housing and Community Development, Baltimore Police Department, and US Census American Community Survey data, 2015-2021

Demolitions are an important tool in Baltimore's arsenal for addressing vacant and abandoned properties, particularly those which are in such disrepair that they pose physical danger to residents (Cohen, 2001). Furthermore, vacancies are known to facilitate crime (Garvin et al., 2013), suggesting that their demolition may directly help in reducing crime and increasing the sense of order for neighborhood residents. However, I find that demolitions happen less often in areas with higher crime. In contrast to the results from my first model, this suggests that these areas are not experiencing the same investment that other areas may see, and which they may need the most.

It is important to compare the findings from the first and second model in relation to how the two types of investments facilitate physical order in communities. Whereas rehabilitations may be happening for homes across a spectrum of conditions, demolitions are primarily used only for those homes which are no longer habitable. Thus, a higher rate of demolitions would be expected in areas with more inhabitable homes. This is supported by the negative correlation between median price of homes sold and the rate of demolitions shown in Figure 2.

Though demolitions may reduce the count of vacant properties, the lot will remain vacant unless new property is built. Vacant lots, though not posing the same dangers of falling buildings, are still costly to the city (Kamau, 2023), and do not necessarily solve the problem of physical disorder and blight. Nevertheless, they are a positive signal to Baltimore residents that action is being taken. Moreover, a lack of demolitions is costly as well. As noted by Kuhlmann (2023), properties which are located near houses that are slated for demolition, but not yet torn down, are more likely to experience a deterioration in their external condition.

My finding of a negative relationship between crime and demolitions, then, suggests that high crime neighborhoods are not experiencing the investment via demolitions from which they

could substantially benefit (in terms of inducing a potential reduction in crime). In other words, these neighborhoods are not granted access to a potential mechanism which may pull them out of blight.

6.4. Prevalence of Unoccupied Residential Properties

Table 4 shows that in areas with higher rates of total crime, the share of unoccupied homes in the following year is significantly higher. If total crime increases by 10%, then in the following year, there are 0.5% more unoccupied homes on average. The sign of the trend is similar for violent crime and property crime, with violent crime being associated with the largest increase in unoccupied homes. Specifically, if violent crime increases by 10%, then the share of unoccupied homes increases by 2.3% on average. If property crime increases 10%, then the share of unoccupied homes increases by 0.6% on average. All results are statistically significant at the 5% level.

Table 4. Relationship between lagged crime and the prevalence of unoccupied homes in a CSA

Crime and Non-Occupancy			
	<i>Dependent variable:</i>		
	Share of Unoccupied Homes		
	(1)	(2)	(3)
Lag of Total Crime	0.054** (0.021)		
Lag of Violent Crime		0.226*** (0.064)	
Lag of Property Crime			0.061** (0.030)
Demographic Controls	Yes	Yes	Yes
Year Fixed Effects	Yes	Yes	Yes
CSA Fixed Effects	Yes	Yes	Yes
# of CSAs	55	55	55
Observations	329	329	329
R ²	0.104	0.125	0.096
Adjusted R ²	-0.152	-0.125	-0.162
<i>Note:</i>	* p<0.1; ** p<0.05; *** p<0.01		

Notes: Standard errors are in parentheses. Levels of statistical significance are indicated at * p<0.1, ** p<0.05, *** p<0.01. Standard errors are in parentheses. 329 observations are used from the full panel of 330 observations due to the one-year lag on crime. Results rendered with stargazer (Hlavac, 2022). Source: Author's calculations using Baltimore Department of Housing and Community Development, Baltimore Police Department, and US Census American Community Survey data, 2015-2021

The third model investigates the relationship between crime and the rate of unoccupied homes as a proxy for social investment. If large shares of homes are unoccupied but still habitable, it is indicative of social disorder more so than physical disorder, as residents are not necessarily forced to move out due to subprime physical conditions, but are instead likely to leave for any variety of social reasons (unfriendly neighbors,(cite?). *Work on this.*

Neighborhoods with social disorder likely have less residential buy-in to social investment.

Indeed, case studies of Baltimore show that residents attribute social disorder to low levels of engagement from neighbors (Cantora et al., 2016). Moreover, Garvin (2013) argues that neighborhood disorder contributes to social isolation, further reducing the sense of collective efficacy (p. 23). In areas with high rates of unoccupied homes, then, social isolation and low levels of engagement are endemic.

The results of my third model show that neighborhoods with higher rates of crime have a higher share of unoccupied homes. This suggests that crime has a discouraging relationship with social investment in a neighborhood. This is troubling for the future of these neighborhoods in Baltimore's attempts to address crime and urban blight, given that the literature strongly asserts that crime happens more in areas where there is less social cohesion and a lower sense of community (Shaw & McKay, 1942; Skogan, 1992; Sampson et al., 1997; Cancino, 2005; Iyer et al., 2020).

It is also notable in this model that the magnitude of the relationship between the variables is larger for violent crime than for property crime. I believe that this provides more support for the idea that unoccupied homes are representative of social disorder more so than physical disorder. If social investment increases less in association with property crime compared to in association with violent crime, then...

Looking at the choropleth maps, there is an obvious similarity in concentrations of unoccupied homes, and concentrations of black residents (Appendices F.6 and F.7).

6.5. Robustness Checks

In Appendix E, I perform a series of robustness checks, including testing for various time-lagged effects of crime, adding additional demographic controls, and a dummy variable to control for the COVID-19 pandemic in 2020 and 2021. The results of these checks affirm the

significance of most results in the primary model. Significance disappears when I remove the lag on crime for the relationship between crime and demolitions (Appendix E.5), and the relationship between total and property crime on non-occupancy (Appendix E.6). Significance also disappears with the two-year lag structure on crime for the relationship to rehabilitations (Appendix E.7). However, these results are in line with the findings of Lacoë et al. (2018), who demonstrate that a one-year lag on crime is the best method for demonstrating the relationship between crime and investment. Looking at current crime does not allow ample time for responses, and looking two years out may be too far removed from any responses.

7. Conclusions

7.1. Policy Recommendations

Recognize that crime and investment likely go hand in hand. In terms of improving neighborhoods/infrastructure/properties...

Increase funding/capacity for demolitions. If they happen less in neighborhoods with higher crime, then they are not effectively solving one of the major implications of a high rate of vacant homes— increased crime. Further, they may be letting crime fester by not demolishing homes and signaling investment to the community.

Something to encourage people to stay in their homes, or to engage with their neighbors. Or maybe investigate why areas with higher crime have more people who aren't receiving mail — are these homes actually unoccupied? Or institute some program to check up on homes that aren't receiving mail — (look to see if other cities do something similar?)

Crime prevention, according to Herbert and Harries (1986), “is less concerned with the search for offenders than with creating conditions in which offenses are less likely to occur” (p. 281). Implication for Baltimore— the methods of crime prevention which rely not on heavily punishing small misdemeanors, but which place more emphasis on remedying the built environment and the community-police relationship, which are currently part of Scott’s plan, will likely be beneficial not just for crime reduction, but also for investment in properties.

Along with that, the City should recognize the potential of area-based policing policies to impact not just crime levels, but the health of their housing market. Continued adaptation of area-based policies.

Relevance of Studying Housing as an Outcome: The Intended Goals of Housing Policy (and Why Crime Prevention Policy Might Help Achieve Those Goals) (from lit review)

As discussed, when neighborhoods are viewed as safe, the value of the homes therein tends to be higher (Tita et al., 2006; Ihlanfeldt & Mayock, 2010; Buonanno et al., 2013; Ceccato & Wilhelmsson, 2020). But is safety the only measure of housing value? And is the sole goal of a housing project to be valued highly? Quality housing should have a positive spillover into many other aspects of a person’s life. If we want to understand the potential that increased safety (via lower crime) has for increased housing development, we should investigate what it is that housing policy is typically attempting to address. In doing so, we can understand the relative importance of reduced crime and increased safety in incentivizing more development.

Frieden (1968) argued that in its most basic form, the problem that housing policy faces is one of “assuring an adequate minimum living environment for people too poor to pay for decent housing at market prices” (p. 583). Today, though, even middle-income families struggle to find adequate housing. Frieden continues by noting that the scale of intervention has

implications for “urban renewal and the future of central cities, metropolitan development, civil rights, and the war on poverty” (ibid). He did note, though, that the basic form of housing policy was insufficient, and that we should expand our view of what housing policy can accomplish. For example, he notes that housing policy should aim to attain the more complex goals of providing freedom of residential choice and the elimination of racial disparities, and should give all families the chance to have access to good schools, adequate public services, and job opportunities (ibid). With this in mind, we can understand contemporary housing policy as attempting to increase access to housing for vulnerable populations, while simultaneously seeking to improve on macro-level trends and increase opportunity for all neighborhoods.

Evaluating new policies with the results in mind

In 2022, Baltimore Mayor Brandon Scott directed \$39 million from federal COVID-19 relief funds to address the crisis of vacant homes plaguing his city. Perhaps unsurprisingly, these funds address metrics that have clear and direct relationships with the housing supply. Specifically, the funds are allocated for building and bulldozing, upgrades to the city’s permitting system, housing enhancements for legacy residents, and more pathways for renters to become homeowners (Miller & Little, 2022). Given the relationship demonstrated in my model between demolitions and crime, it would be useful for officials in Baltimore to evaluate the performance of this policy in regards to how it affects the crime rate. If crime decreases in areas where revitalization is occurring, then this could be promising for the overall outlook for urban decay in Baltimore.

(Crime Policies)

Harrison's plan is about reforming the department as much as it is about violence reduction across the city. Similarly, Scott's plan is as much about reforming the city's approach to crime and violence, as it is about actually reducing that violence. Both plans have a general outline for framing their objectives.

Harrison's plan has the following main objectives: 1) reducing crime with comprehensive, data-backed strategies; 2) strengthening partnerships; 3) transforming the police department via the consent decree; and 4) continuing that transformation via a renewed focus on recruitment and retention.

Scott's plan revolves around three "pillars," each with their own desired outcomes and impacts: 1) public health approach to violence, 2) community engagement and inter-agency coordination, and 3) evaluation and accountability

(Moving past broken windows) Ralph Taylor (2001), a critic of the broken windows theory, argues that certain characteristics such as litter and rowdy teenagers "may be better interpreted as a result of an economically disadvantaged neighborhood rather than as ...symptom[s] of a disorderly or disorganized neighborhood" (Lombardo & Lough, 2007, p. 126). Accordingly, some efforts to fix the problem of "broken windows" have focused on providing economic support rather than on punishing people. After all, as Taylor points out, the data is unclear as to whether social or physical disorder has a greater effect on serious crime. If there is indeed no strong link between social disorder and crime, given that there is no strong reliable data, then it may be wiser from a policy standpoint to invest in rehabilitating *physical* disorder, rather than trying to quell *social* disorder (Collins, p. 426). .

Debates and Controversies in Crime Prevention Strategies (from lit review)

Hence, the net benefits of broken windows policing are often debated. Scholars point out the theory's lack of strong empirical support and claim that any potential benefits cannot possibly be enough to account for the harms its tactics can inflict (Collins, 2007, p. 423). For example, Bernard Harcourt (2001) points out the lack of strong controls and the over-application of studies such as Skogan's (1990) and Sampson and Cohen (1988), pointing out that their conclusions were essentially sensationalized and misconstrued to support the continued use of broken windows tactics (Collins, 2007, p. 424).

One aspect of the broken windows practice that particularly concerns scholars is its disproportionate effect on people of color, especially black people. In Baltimore specifically, the DOJ's 2016 investigation confirmed that black people consistently made up a disproportionate share of arrests or charges made for minor offenses. For example, blacks made up over 80% of the charges made in the following violations: resisting arrest (87%), making a false statement (89%), failing to obey an order (84%), disorderly conduct (83%), and trespassing (88%) (Graham, 2016). For context, black people constituted about 61% of the city's population at the time (Census). The spread between the share of blacks being arrested for these procedural misdemeanors and their actual population is indicative of a common problem in crime prevention—the use of race as a proxy for criminal attributes. Moreover, scholars argue that these methods have contributed to a delegitimization of the police force by eroding the public's trust in the police force's commitment to ensuring due process, especially for minority groups (Kamalu & Onyeozili, 2018). **Thus, some reforms in crime prevention address this problem by expanding opportunities for community engagement. These approaches aim to change**

the focus of policing— from fighting crime to *preventing* crime, by working with the community to address issues of safety and trust in law enforcement (Baltimore City Comprehensive Violence Prevention Plan, 2021, p. 18)

The result of this transformation in methods is what I will be discussing next— the **methods of crime prevention which rely not on heavily punishing small misdemeanors, but which place more emphasis on remedying the built environment and the community-police relationship.** ~~These are also the principles on which the Baltimore policing transformation and violence prevention plans are based— prioritizing public health, harm reduction, and community input and capacity building.~~

Rethinking Crime Prevention: Area-Based Policies

The literature presents two major branches of the criminal justice system: law enforcement and crime control. Whereas law enforcement is concerned with bringing to justice those who break the law, crime control is a broad range of activities designed to increase citizen respect and recognition of the law (Morris & Heal, 1981; Herbert and Harries, 1986, p. 284). Thus, we might understand broken windows policing as a branch of law enforcement, whereas the strategies I will describe below align more closely with practices of crime control.

Indeed, as observed by McGarrell et al. (1999), there has been a persistent trend since the late 20th century of efforts to address crime at the neighborhood level. Critically, these developments have been indicative of “a steady and growing recognition that the police and the citizenry are on the front line of this battle [against crime] and must do more than just react to the problem after the fact” (Rosenbaum, 1986, p. 11).

Environmental Concerns

It is worth noting that recent debates about housing development have brought up the importance of environmental sustainability, especially as it relates to compact redevelopment (Mueller & Steiner, 2010). While important issues, they extend beyond the reach of this paper, which intends to study the relationship housing development has with crime. As such, I will not be discussing these topics at length, but encourage future research to consider the impact of crime rates and crime policies on the uptake of environmentally sustainable housing development.

7.2. Limitations

Crime Data

Crime statistics are officially reported by the Baltimore Police Department. However, crime data has its limitations. Reflecting just those incidents which are reported and properly logged by police officers, official crime statistics may deviate from perceptions of crime in a community (Buonanno et al., 2013). My model does not capture the differential relation to investment that reported crime may have as compared to perceptions of crime. This is a limitation of crime studies which is well documented and discussed (MacDonald, 2002; Gibbons, 2004; Tita et al. 2006; Buonanno et al., 2013). However, in lieu of having the capacity to track crime that goes unreported, this data serves as the current most accessible measure of crime in Baltimore. Future research should seek to isolate this difference, perhaps by using victimization data rather than reported crime rates (Buonanno et al., 2013).

Size of CSAs

CSAs are larger than census tracts and neighborhoods—they consist of 1-8 tracts and have populations of up to 20,000 people (Appendix A). Accordingly, crime that happens on one side of a CSA may not impact residents living at the other end to the same degree. Future studies of Baltimore may investigate this potential limitation by matching the housing indicator data to the corresponding census tracts, and running the same models to see if the relationship between crime and investment changes at a more granular geographic level.

That being said, some CSAs only have a few census tracts (Appendix B). Therefore, the differences between the units may not be large enough to result in significantly different findings.

Availability of Data

Data on community vital signs is only available from 2015 to 2021. With more data, results and conclusions would be more robust. On the other hand, though, including more data would necessitate controls for events such as the Great Recession of 2007-2009, which may have differentially impacted families and the indicators relevant to their housing investment, such as employment and levels of disposable income that may be used for rehabilitations.

Crime Displacement

Though rehabilitation and investment in properties is promising for its relationship with lower crime rates, there is a limitation of this approach for reducing crime wholesale. There is the possibility that crime does not decrease overall, but merely shifts locations (i.e. the crime is “displaced”). In 1999, Schumacher and Leitner used a hot spot analysis to test if this is the case in Baltimore, by looking at whether urban renewal programs in Baltimore have affected the spatial distribution of crime. They find that the projects with the most potential to discourage

criminal activity are: increased presence of security personnel, improved street lighting, and increased pedestrian traffic. However, they note that the *overall* crime rate in the city remained constant. They conclude, then, that these interventions do not eliminate criminal activity— they merely displace it.

I recommend that future research investigate the possibility of displacement as it relates to the current crime prevention policies in Baltimore. However, given that crime data indeed shows an overall decrease in Baltimore over the past few years, this may not be an urgent issue.

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Appendices

Appendix A: Community Statistical Areas

The Baltimore Neighborhood Indicators Alliance Jacob France Institute (BNIA-JFI) uses Community Statistical Areas (CSAs) to consolidate data on over 270 neighborhoods in Baltimore. Neighborhoods are small areas combining social and geographic aspects, but the boundaries defined by the city can differ from those understood by residents. Alternatively, CSAs are consistent boundaries, allowing more accurate comparisons of conditions over time.

CSAs were originally developed by the Baltimore Data Collaborative with the Baltimore City Department of Planning, using four guidelines:

1. CSA boundaries had to align with Census Tracts;

2. CSAs would consist of 1-8 tracts, preferably with total populations in the range of 5,000 to 20,000;
3. CSAs would define relatively demographically homogeneous areas;
4. CSAs should reflect the City planners' understanding of residents' and institutions' perceptions of the boundaries of the community (*BNIA Mapping Resources*, n.d.)

With the results of the decennial census, census tract definitions may change. Accordingly, BNIA-JFI may re-evaluate the CSA boundaries. Given the results of the 2020 census, however, BNIA-JFI decided to maintain the existing CSA boundaries, only changing the names of some areas at the request of community members. Thus, my unit of analysis remained constant from pre-2020 to post-2020.

Figure A.1. Community Statistical Areas in Baltimore (2010)



Source: Baltimore City Health Department.

Appendix B: Map of Baltimore CSAs and Census Tracts



Source: Baltimore Neighborhood Indicators Alliance

Appendix C: Modifications of Variables in the Data

C.1. Formula for adjusting prices to 2021 dollars

$$\text{Current value} = \text{Original value} \times \left(\frac{\text{Current CPI}}{\text{CPI in the past}} \right).$$

Source: St. Louis Fed, <https://research.stlouisfed.org/publications/page1-econ/2023/01/03/adjusting-for-inflation>

C.2. Real Median Household Income

```
# real median hh income in 2021 dollars
real_median_hh_income <- median_hh_income %>%
  mutate(Real_Income15 = mghi15 * (271/237),
         Real_Income16 = mghi16 * (271/240),
         Real_Income17 = mghi17 * (271/245.1),
         Real_Income18 = mghi18 * (271/251.1),
         Real_Income19 = mghi19 * (271/255.7),
         Real_Income20 = mghi20 * (271/258.8),
         Real_Income21 = mghi21)
```

C.3. Real Median Price of Homes Sold

```
# real prices in 2021 dollars
real_median_price_sold <- median_price_sold %>%
  mutate(Real_Price_Sold15 = salepr15 * (271/237),
         Real_Price_Sold16 = salepr16 * (271/240),
         Real_Price_Sold17 = salepr17 * (271/245.1),
         Real_Price_Sold18 = salepr18 * (271/251.1),
         Real_Price_Sold19 = salepr19 * (271/255.7),
         )
```

C.3. Total Crime Rate per CSA Capita

```
# calculate crime rate per csa capita (wide-form)
total_crimer <- total_crime %>% rename(Community = CSA2010) %>%
  left_join(csa_population, by = "Community") %>% # join CSA population data
  mutate(crimer15 = (`crime15`*1000)/Population,
         crimer16 = (`crime16`*1000)/Population,
         crimer17 = (`crime17`*1000)/Population,
         crimer18 = (`crime18`*1000)/Population,
         crimer19 = (`crime19`*1000)/Population,
         crimer20 = (`crime20`*1000)/Population,
         crimer21 = (`crime21`*1000)/Population)

# check data
total_crimer
# tidy to long form
total_crimer_long <- pivot_long(total_crimer, "crimer", "total_crimer")
# display data
total_crimer_long # crime rate per csa capita
```

C.4. Violent Crime Rate per CSA Capita

```
# calculate violent crime rate per csa capita
vio_crimer <- vio_crime %>% rename(Community = CSA2010) %>%
  left_join(csa_population, by = "Community") %>% # join CSA population data
  mutate(vio_crimer15 = (`viol15`*1000)/Population,
         vio_crimer16 = (`viol16`*1000)/Population,
         vio_crimer17 = (`viol17`*1000)/Population,
         vio_crimer18 = (`viol18`*1000)/Population,
         vio_crimer19 = (`viol19`*1000)/Population,
         vio_crimer20 = (`viol20`*1000)/Population,
         vio_crimer21 = (`viol21`*1000)/Population)

# check
vio_crimer
# tidy to long form
vio_crimer_long <- pivot_long(vio_crimer, "vio_crimer", "vio_crimer")
# check
vio_crimer_long # violent crime rate per CSA capita
```

C.5. Property Crime Rate per CSA Capita

```
# calculate property crime rate by CSA pop
prop_crime <- left_join(as.data.frame(population_csa), wide_propcrime,
  by = c("CSA2010" = "Community")) %>% # join CSA population data
  transmute(CSA2010 = CSA2010,
            crimer15 = (`2015`*1000)/Population,
            crimer16 = (`2016`*1000)/Population,
            crimer17 = (`2017`*1000)/Population,
            crimer18 = (`2018`*1000)/Population,
            crimer19 = (`2019`*1000)/Population,
            crimer20 = (`2020`*1000)/Population,
            crimer21 = (`2021`*1000)/Population)

# check
prop_crime
# tidy to long form
prop_crimer_long <- pivot_long(prop_crime, "crimer", "prop_crimer")
prop_crimer_long
```

Appendix D: Regression Outputs

D.1. Crime and Rehabilitations, expanded control variables

Full OLS Regression Results: Rehabilitations			
	<i>Dependent variable:</i>		
	Rate of Rehabilitation Permits		
	(1)	(2)	(3)
Lag of Total Crime	0.050** (0.023)		
Lag of Violent Crime		0.144** (0.069)	
Lag of Property Crime			0.057* (0.032)
Black	-0.098*** (0.034)	-0.094*** (0.034)	-0.101*** (0.034)
Asian	-0.105 (0.074)	-0.107 (0.075)	-0.094 (0.074)
Hispanic	-0.059* (0.034)	-0.055 (0.034)	-0.060* (0.034)
Two or more races	-0.044 (0.090)	-0.051 (0.090)	-0.044 (0.090)
All other races	-0.432*** (0.141)	-0.420*** (0.141)	-0.425*** (0.141)
Under 5 years	-0.083 (0.075)	-0.078 (0.075)	-0.082 (0.076)
5-17 years	0.012 (0.070)	0.010 (0.070)	0.014 (0.070)
18-24 years	-0.085 (0.072)	-0.085 (0.073)	-0.086 (0.073)
25-64 years	-0.043 (0.057)	-0.040 (0.057)	-0.041 (0.058)
HS and some College	0.034 (0.041)	0.032 (0.041)	0.033 (0.041)
Bachelor's or more	0.0001 (0.043)	0.0002 (0.043)	-0.001 (0.044)
Employed	0.028 (0.030)	0.028 (0.030)	0.027 (0.030)
Below Poverty	-0.009 (0.030)	-0.007 (0.030)	-0.011 (0.030)
Year Fixed Effects	Yes	Yes	Yes
CSA Fixed Effects	Yes	Yes	Yes
# of CSAs	55	55	55
Observations	384	384	384
R ²	0.084	0.082	0.079
Adjusted R ²	-0.136	-0.137	-0.142
Note:	* p<0.1; ** p<0.05; *** p<0.01		

Notes: Standard errors are in parentheses. Levels of statistical significance are indicated at * p<0.1, ** p<0.05, *** p<0.01. Standard errors are in parentheses. 384 observations are used from the full panel of 385 observations due to the one-year lag on crime. Source: Author's calculations using Baltimore Department of Housing and Community Development, Baltimore Police Department, and US Census American Community Survey data, 2015-2021

D.2. Crime and Demolitions, expanded control variables

Full OLS Regression Results: Demolitions			
	Dependent variable:		
	Rate of Demolition Permits		
	(1)	(2)	(3)
Lag of Total Crime	-0.205*** (0.077)		
Lag of Violent Crime		-0.676*** (0.232)	
Lag of Property Crime			-0.245** (0.109)
Black	0.066 (0.114)	0.046 (0.114)	0.076 (0.115)
Asian	0.208 (0.251)	0.233 (0.251)	0.167 (0.250)
Hispanic	0.013 (0.115)	-0.0004 (0.114)	0.018 (0.115)
Two or more races	-0.375 (0.302)	-0.340 (0.302)	-0.374 (0.303)
All other races	0.352 (0.474)	0.302 (0.473)	0.326 (0.475)
Under 5 years	-0.189 (0.254)	-0.200 (0.253)	-0.191 (0.255)
5-17 years	0.066 (0.235)	0.076 (0.234)	0.059 (0.236)
18-24 years	-0.038 (0.244)	-0.040 (0.244)	-0.035 (0.245)
25-64 years	-0.025 (0.194)	-0.032 (0.193)	-0.031 (0.194)
HS and some College	-0.123 (0.138)	-0.114 (0.138)	-0.121 (0.139)
Bachelor's or more	-0.019 (0.146)	-0.021 (0.146)	-0.015 (0.147)
Employed	-0.029 (0.101)	-0.029 (0.100)	-0.025 (0.101)
Below Poverty	0.026 (0.100)	0.015 (0.100)	0.034 (0.100)
Year Fixed Effects	Yes	Yes	Yes
CSA Fixed Effects	Yes	Yes	Yes
# of CSAs	55	55	55
Observations	384	384	384
R ²	0.043	0.047	0.037
Adjusted R ²	-0.187	-0.181	-0.194
Note:	* p<0.1; ** p<0.05; *** p<0.01		

Notes: Standard errors are in parentheses. Levels of statistical significance are indicated at * p<0.1, ** p<0.05, *** p<0.01. Standard errors are in parentheses. 384 observations are used from the full panel of 385 observations due to the one-year lag on crime. Source: Author's calculations using Baltimore Department of Housing and Community Development, Baltimore Police Department, and US Census American Community Survey data, 2015-2021

D.3. Crime and Non-Occupancy, expanded control variables

Full OLS Regression Results: Non-Occupancy			
	<i>Dependent variable:</i>		
	Share of Unoccupied Homes		
	(1)	(2)	(3)
Lag of Total Crime	0.054** (0.021)		
Lag of Violent Crime		0.226*** (0.064)	
Lag of Property Crime			0.061** (0.030)
Black	-0.096*** (0.037)	-0.090** (0.036)	-0.098*** (0.037)
Asian	0.018 (0.077)	0.004 (0.076)	0.029 (0.077)
Hispanic	-0.055 (0.035)	-0.050 (0.035)	-0.057 (0.036)
Two or more races	-0.092 (0.113)	-0.103 (0.112)	-0.093 (0.114)
All other races	-0.191 (0.149)	-0.179 (0.148)	-0.184 (0.150)
Under 5 years	0.004 (0.083)	0.001 (0.082)	0.005 (0.084)
5-17 years	0.057 (0.074)	0.049 (0.073)	0.060 (0.074)
18-24 years	0.048 (0.075)	0.053 (0.074)	0.046 (0.076)
25-64 years	-0.010 (0.060)	-0.013 (0.059)	-0.006 (0.060)
HS and some College	-0.037 (0.044)	-0.040 (0.044)	-0.037 (0.045)
Bachelor's or more	-0.083* (0.047)	-0.085* (0.046)	-0.082* (0.047)
Employed	-0.022 (0.032)	-0.024 (0.032)	-0.022 (0.033)
Below Poverty	0.079*** (0.030)	0.084*** (0.029)	0.078*** (0.030)
Year Fixed Effects	Yes	Yes	Yes
CSA Fixed Effects	Yes	Yes	Yes
# of CSAs	55	55	55
Observations	329	329	329
R ²	0.104	0.125	0.096
Adjusted R ²	-0.152	-0.125	-0.162
<i>Note:</i>	* p<0.1; ** p<0.05; *** p<0.01		

Notes: Standard errors are in parentheses. Levels of statistical significance are indicated at * p<0.1, ** p<0.05, *** p<0.01. Standard errors are in parentheses. 384 observations are used from the full panel of 385 observations due to the one-year lag on crime. Source: Author's calculations using Baltimore Department of Housing and Community Development, Baltimore Police Department, and US Census American Community Survey data, 2015-2021

Appendix E: Robustness Checks

The robustness checks in the following sections include: adding a COVID-19 dummy variable for years 2020 and 2021, using explanatory variables of interest in the current time period instead of the one-year lagged period, using explanatory variables of interest in the two year-lagged time period instead of the one-year lagged period, and replacing the shares of the population by educational attainment with the log of median household income as a continuous variable.

E.1. Including a COVID Dummy on the Original Two-Way Fixed Effects Regression of Rehabilitation Permits on 1-year Crime Lag

	<i>Dependent variable:</i>		
	Rate of Rehabilitation Permits		
	(1)	(2)	(3)
Lag of Total Crime	0.050** (0.023)		
Lag of Violent Crime		0.144** (0.069)	
Lag of Property Crime			0.057* (0.032)
Demographic Controls	Yes	Yes	Yes
COVID Dummy	Yes	Yes	Yes
Year Fixed Effects	Yes	Yes	Yes
CSA Fixed Effects	Yes	Yes	Yes
# of CSAs	55	55	55
Observations	384	384	384
R ²	0.084	0.082	0.079
Adjusted R ²	-0.136	-0.137	-0.142
<i>Note:</i>	* p<0.1; ** p<0.05; *** p<0.01		

Notes: Standard errors are in parentheses. Levels of statistical significance are indicated at * p<0.1, ** p<0.05, *** p<0.01. 384 observations are used from the full panel of 385 observations due to the one-year lag on crime. Source: Author's calculations using Baltimore Department of Housing and Community Development, Baltimore Police Department, and US Census American Community Survey data, 2015-2021

E.2. Including a COVID Dummy on the Original Two-Way Fixed Effects Regression of Demolition Permits on 1-year Crime Lag

Crime and Demolitions			
	<i>Dependent variable:</i>		
	Rate of Demolition Permits		
	(1)	(2)	(3)
Lag of Total Crime	-0.205*** (0.077)		
Lag of Violent Crime		-0.676*** (0.232)	
Lag of Property Crime			-0.245** (0.109)
Demographic Controls	Yes	Yes	Yes
COVID Dummy	Yes	Yes	Yes
Year Fixed Effects	Yes	Yes	Yes
CSA Fixed Effects	Yes	Yes	Yes
# of CSAs	55	55	55
Observations	384	384	384
R ²	0.043	0.047	0.037
Adjusted R ²	-0.187	-0.181	-0.194
<i>Note:</i>	* p<0.1; ** p<0.05; *** p<0.01		

Notes: Standard errors are in parentheses. Levels of statistical significance are indicated at * p<0.1, ** p<0.05, *** p<0.01. 384 observations are used from the full panel of 385 observations due to the one-year lag on crime. Source: Author's calculations using Baltimore Department of Housing and Community Development, Baltimore Police Department, and US Census American Community Survey data, 2015-2021

E.3. Including a COVID Dummy on the Original Two-Way Fixed Effects Regression of Share of Unoccupied Homes on 1-year Crime Lag

Crime and Non-Occupancy			
	<i>Dependent variable:</i>		
	Share of Unoccupied Homes		
	(1)	(2)	(3)
Lag of Total Crime	0.054** (0.021)		
Lag of Violent Crime		0.226*** (0.064)	
Lag of Property Crime			0.061** (0.030)
Demographic Controls	Yes	Yes	Yes
COVID Dummy	Yes	Yes	Yes
Year Fixed Effects	Yes	Yes	Yes
CSA Fixed Effects	Yes	Yes	Yes
# of CSAs	55	55	55
Observations	329	329	329
R ²	0.104	0.125	0.096
Adjusted R ²	-0.152	-0.125	-0.162
<i>Note:</i>	* p<0.1; ** p<0.05; *** p<0.01		

Notes: Standard errors are in parentheses. Levels of statistical significance are indicated at * p<0.1, ** p<0.05, *** p<0.01. 329 observations are used from the full panel of 330 observations due to the one-year lag on crime. Source: Author's calculations using Baltimore Department of Housing and Community Development, Baltimore Police Department, and US Census American Community Survey data, 2015-2021

E.4. Modifying the Original Two-Way Fixed Effects Regression with Crime Rate in Current Period; Results for Prevalence of Rehabilitation Permits

Crime and Rehabilitation Permits			
	<i>Dependent variable:</i>		
	Rate of Rehabilitation Permits		
	(1)	(2)	(3)
Total Crime	0.205*** (0.045)		
Violent Crime		0.519*** (0.136)	
Property Crime			0.213*** (0.054)
Demographic Controls	Yes	Yes	Yes
Year Fixed Effects	Yes	Yes	Yes
CSA Fixed Effects	Yes	Yes	Yes
# of CSAs	55	55	55
Observations	385	385	385
R ²	0.129	0.111	0.113
Adjusted R ²	-0.079	-0.102	-0.099
<i>Note:</i>	* p<0.1; ** p<0.05; *** p<0.01		

Notes: Standard errors are in parentheses. Levels of statistical significance are indicated at * p<0.1, ** p<0.05, *** p<0.01. Source: Author's calculations using Baltimore Department of Housing and Community Development, Baltimore Police Department, and US Census American Community Survey data, 2015-2021

E.5. Modifying the Original Two-Way Fixed Effects Regression Crime Rate in Current Period; Results for Prevalence of Demolition Permits

Crime and Demolitions			
	<i>Dependent variable:</i>		
	Rate of Demolition Permits		
	(1)	(2)	(3)
Total Crime	-0.106 (0.155)		
Violent Crime		-0.860* (0.469)	
Property Crime			-0.078 (0.189)
Demographic Controls	Yes	Yes	Yes
Year Fixed Effects	Yes	Yes	Yes
CSA Fixed Effects	Yes	Yes	Yes
# of CSAs	55	55	55
Observations	385	385	385
R ²	0.022	0.031	0.021
Adjusted R ²	-0.211	-0.200	-0.212
<i>Note:</i>	* p<0.1; ** p<0.05; *** p<0.01		

Notes: Standard errors are in parentheses. Levels of statistical significance are indicated at * p<0.1, ** p<0.05, *** p<0.01. Source: Author's calculations using Baltimore Department of Housing and Community Development, Baltimore Police Department, and US Census American Community Survey data, 2015-2021

E.6. Modifying the Original Two-Way Fixed Effects Regression with Crime Rate in Current Period; Results for Share of Unoccupied Homes

Crime and Non-Occupancy			
	<i>Dependent variable:</i>		
	Share of Unoccupied Homes		
	(1)	(2)	(3)
Total Crime	0.051 (0.047)		
Violent Crime		0.334** (0.142)	
Property Crime			0.025 (0.055)
Demographic Controls	Yes	Yes	Yes
Year Fixed Effects	Yes	Yes	Yes
CSA Fixed Effects	Yes	Yes	Yes
# of CSAs	55	55	55
Observations	330	330	330
R ²	0.086	0.101	0.082
Adjusted R ²	-0.175	-0.155	-0.179
<i>Note:</i>	* p<0.1; ** p<0.05; *** p<0.01		

Notes: Standard errors are in parentheses. Levels of statistical significance are indicated at * p<0.1, ** p<0.05, *** p<0.01. Source: Author's calculations using Baltimore Department of Housing and Community Development, Baltimore Police Department, and US Census American Community Survey data, 2015-2021

E.7. Modifying the Original Two-Way Fixed Effects Regression with a 2-year Lagged Crime Rate; Results for Prevalence of Rehabilitation Permits

Crime and Rehabilitation Permits			
	<i>Dependent variable:</i>		
	Rate of Rehabilitation Permits		
	(1)	(2)	(3)
2-Year Lag, Total Crime	0.012 (0.019)		
2-Year Lag, Violent Crime		0.033 (0.058)	
2-Year Lag, Property Crime			0.003 (0.028)
Demographic Controls	Yes	Yes	Yes
Year Fixed Effects	Yes	Yes	Yes
CSA Fixed Effects	Yes	Yes	Yes
# of CSAs	55	55	55
Observations	383	383	383
R ²	0.071	0.070	0.070
Adjusted R ²	-0.153	-0.153	-0.154
<i>Note:</i>	* p<0.1; ** p<0.05; *** p<0.01		

Notes: Standard errors are in parentheses. Levels of statistical significance are indicated at * p<0.1, ** p<0.05, *** p<0.01. Standard errors are in parentheses. 383 observations are used from the full panel of 385 observations due to the two-year lag on crime. Source: Author's calculations using Baltimore Department of Housing and Community Development, Baltimore Police Department, and US Census American Community Survey data, 2015-2021

E.8. Modifying the Original Two-Way Fixed Effects Regression with a 2-year Lagged Crime Rate; Results for Prevalence of Demolition Permits

Crime and Demolition Permits			
	<i>Dependent variable:</i>		
	Rate of Demolition Permits		
	(1)	(2)	(3)
2-Year Lag, Total Crime	-0.199*** (0.064)		
2-Year Lag, Violent Crime		-0.648*** (0.193)	
2-Year Lag, Property Crime			-0.259*** (0.092)
Demographic Controls	Yes	Yes	Yes
Year Fixed Effects	Yes	Yes	Yes
CSA Fixed Effects	Yes	Yes	Yes
# of CSAs	55	55	55
Observations	383	383	383
R ²	0.051	0.056	0.045
Adjusted R ²	-0.178	-0.171	-0.184
<i>Note:</i>	* p<0.1; ** p<0.05; *** p<0.01		

Notes: Standard errors are in parentheses. Levels of statistical significance are indicated at * p<0.1, ** p<0.05, *** p<0.01. Standard errors are in parentheses. 383 observations are used from the full panel of 385 observations due to the two-year lag on crime. Source: Author's calculations using Baltimore Department of Housing and Community Development, Baltimore Police Department, and US Census American Community Survey data, 2015-2021

E.9. Modifying the Original Two-Way Fixed Effects Regression with a 2-year Lagged Crime Rate; Results for Share of Unoccupied Homes

Crime and Non-Occupancy			
	<i>Dependent variable:</i>		
	Share of Unoccupied Homes		
	(1)	(2)	(3)
2-Year Lag, Total Crime	-0.232*** (0.068)		
2-Year Lag, Violent Crime		-0.711*** (0.205)	
2-Year Lag, Property Crime			-0.320*** (0.098)
Demographic Controls	Yes	Yes	Yes
Year Fixed Effects	Yes	Yes	Yes
CSA Fixed Effects	Yes	Yes	Yes
# of CSAs	55	55	55
Observations	328	328	328
R ²	0.069	0.070	0.065
Adjusted R ²	-0.199	-0.197	-0.203
<i>Note:</i>	* p<0.1; ** p<0.05; *** p<0.01		

Notes: Standard errors are in parentheses. Levels of statistical significance are indicated at * p<0.1, ** p<0.05, *** p<0.01. 383 observations are used from the full panel of 385 observations due to the two-year lag on crime. Source: Author's calculations using Baltimore Department of Housing and Community Development, Baltimore Police Department, and US Census American Community Survey data, 2015-2021

E.10. Replacing Education Controls with Median Household Income Control; Results for Prevalence of Rehabilitation Permits

Crime and Rehabilitation Permits			
	<i>Dependent variable:</i>		
	Rate of Rehabilitation Permits		
	(1)	(2)	(3)
Lag of Total Crime	0.050** (0.023)		
Lag of Violent Crime		0.146** (0.069)	
Lag of Property Crime			0.057* (0.032)
Demographic Controls	Yes	Yes	Yes
Log of Median Household Income	Yes	Yes	Yes
Year Fixed Effects	Yes	Yes	Yes
CSA Fixed Effects	Yes	Yes	Yes
# of CSAs	55	55	55
Observations	384	384	384
R ²	0.080	0.079	0.075
Adjusted R ²	-0.136	-0.138	-0.143
<i>Note:</i>	* p<0.1; ** p<0.05; *** p<0.01		

Notes: Standard errors are in parentheses. Levels of statistical significance are indicated at * p<0.1, ** p<0.05, *** p<0.01. 384 observations are used from the full panel of 385 observations due to the one-year lag on crime. Source: Author's calculations using Baltimore Department of Housing and Community Development, Baltimore Police Department, and US Census American Community Survey data, 2015-2021

E.11. Replacing Education Controls with Median Household Income Control; Results for Prevalence of Demolition Permits

Crime and Demolition Permits			
	<i>Dependent variable:</i>		
	Rate of Demolition Permits		
	(1)	(2)	(3)
Lag of Total Crime	-0.199*** (0.075)		
Lag of Violent Crime		-0.651*** (0.227)	
Lag of Property Crime			-0.241** (0.107)
Demographic Controls	Yes	Yes	Yes
Log of Median Household Income	Yes	Yes	Yes
Year Fixed Effects	Yes	Yes	Yes
CSA Fixed Effects	Yes	Yes	Yes
# of CSAs	55	55	55
Observations	384	384	384
R ²	0.082	0.085	0.076
Adjusted R ²	-0.135	-0.130	-0.141
<i>Note:</i>	* p<0.1; ** p<0.05; *** p<0.01		

Notes: Standard errors are in parentheses. Levels of statistical significance are indicated at * p<0.1, ** p<0.05, *** p<0.01. 384 observations are used from the full panel of 385 observations due to the one-year lag on crime. Source: Author's calculations using Baltimore Department of Housing and Community Development, Baltimore Police Department, and US Census American Community Survey data, 2015-2021

E.12. Replacing Education Controls with Median Household Income Control; Results for Share of Unoccupied Homes

Crime and Non-Occupancy			
	<i>Dependent variable:</i>		
	Share of Unoccupied Homes		
	(1)	(2)	(3)
Lag of Total Crime	0.055** (0.021)		
Lag of Violent Crime		0.227*** (0.064)	
Lag of Property Crime			0.063** (0.030)
Demographic Controls	Yes	Yes	Yes
Log of Median Household Income	Yes	Yes	Yes
Year Fixed Effects	Yes	Yes	Yes
CSA Fixed Effects	Yes	Yes	Yes
# of CSAs	55	55	55
Observations	329	329	329
R ²	0.092	0.113	0.084
Adjusted R ²	-0.163	-0.137	-0.173
<i>Note:</i>	* p<0.1; ** p<0.05; *** p<0.01		

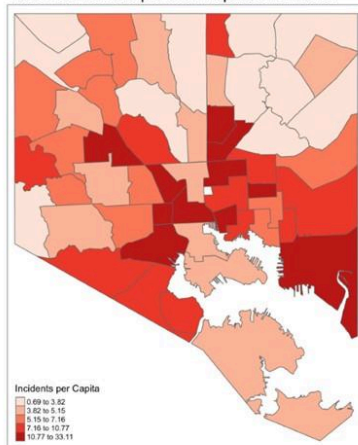
Notes: Standard errors are in parentheses. Levels of statistical significance are indicated at * p<0.1, ** p<0.05, *** p<0.01. 329 observations are used from the full panel of 330 observations due to the one-year lag on crime. Source: Author's calculations using Baltimore Department of Housing and Community Development, Baltimore Police Department, and US Census American Community Survey data, 2015-2021

Appendix F: Maps

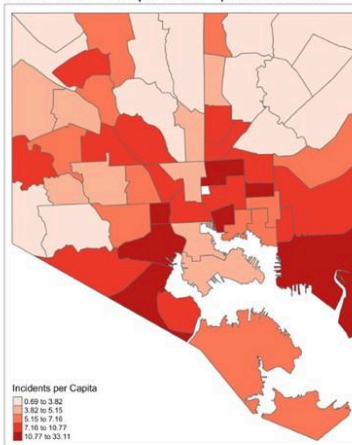
F.1. Part 1 Crime

APPENDIX F.1: PART 1: CRIME RATE PER CSA POPULATION

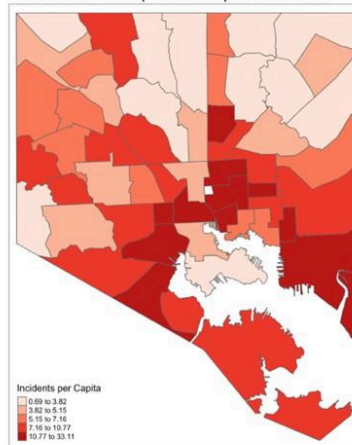
Part 1 Crime Rate per CSA Population in 2015



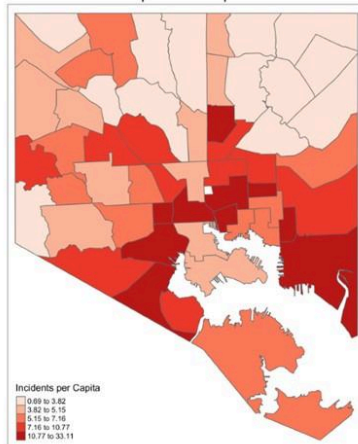
Part 1 Crime Rate per CSA Population in 2016



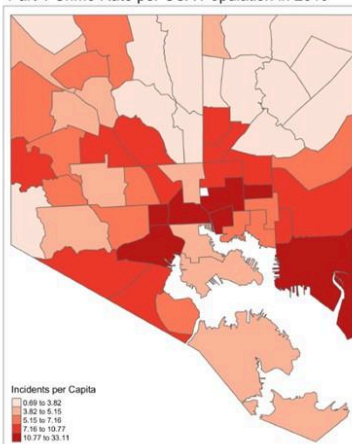
Part 1 Crime Rate per CSA Population in 2017



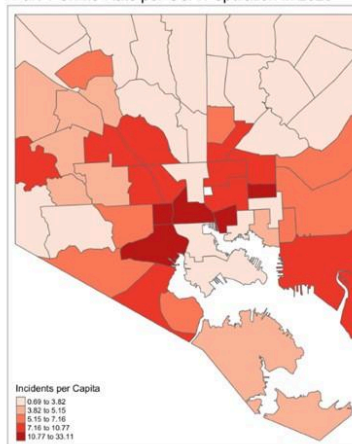
Part 1 Crime Rate per CSA Population in 2018



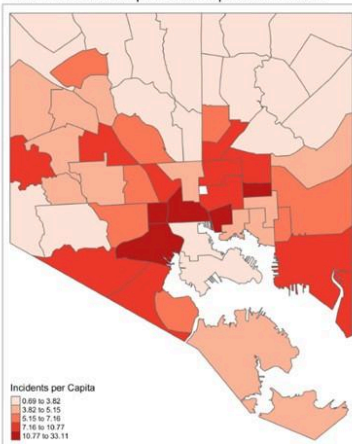
Part 1 Crime Rate per CSA Population in 2019



Part 1 Crime Rate per CSA Population in 2020

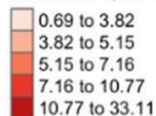


Part 1 Crime Rate per CSA Population in 2021



LEGEND

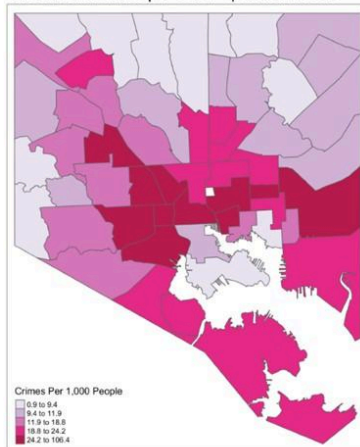
Incidents per Capita



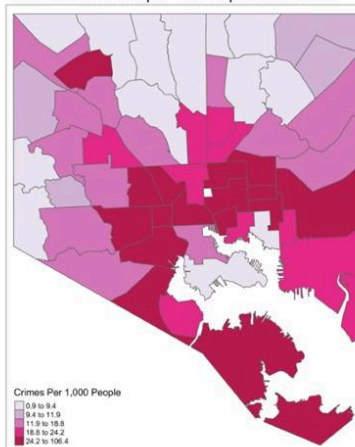
F.2. Violent Crime

APPENDIX F.2: VIOLENT CRIME PER CSA POPULATION

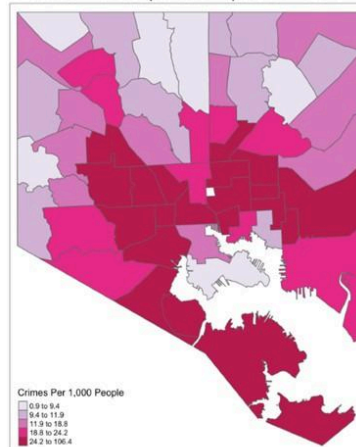
Violent Crime Rate per CSA Population in 2015



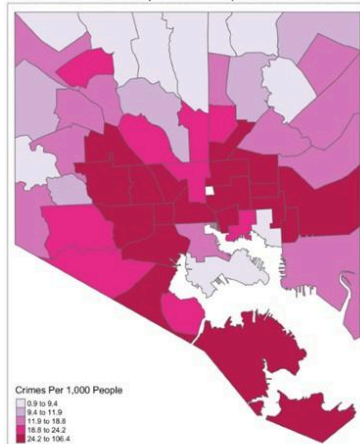
Violent Crime Rate per CSA Population in 2016



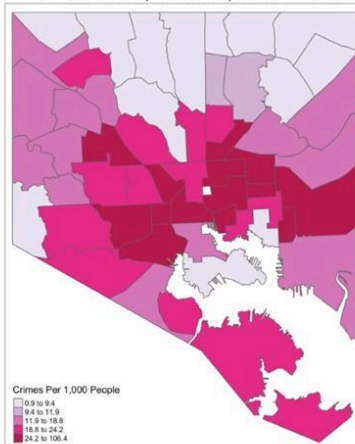
Violent Crime Rate per CSA Population in 2017



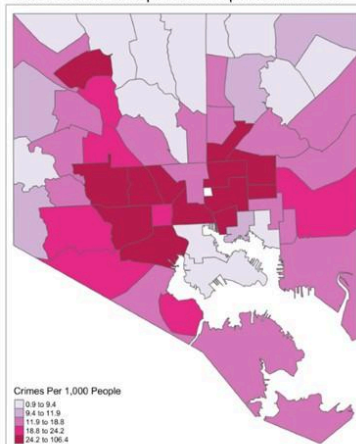
Violent Crime Rate per CSA Population in 2018



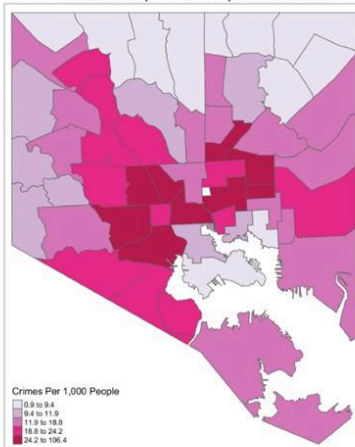
Violent Crime Rate per CSA Population in 2019



Violent Crime Rate per CSA Population in 2020

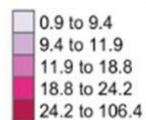


Violent Crime Rate per CSA Population in 2021



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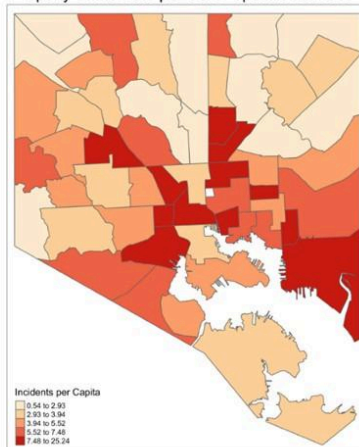
Crimes Per 1,000 People



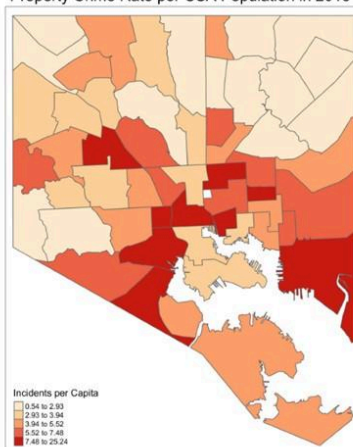
F.3. Property Crime

APPENDIX F.3: PROPERTY CRIME RATE PER CSA POPULATION

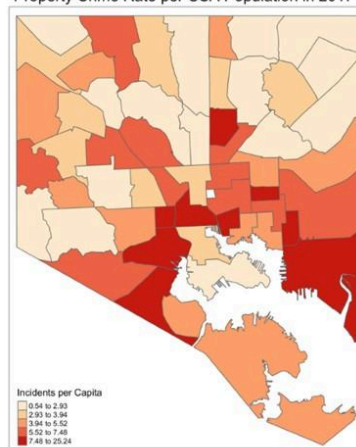
Property Crime Rate per CSA Population in 2015



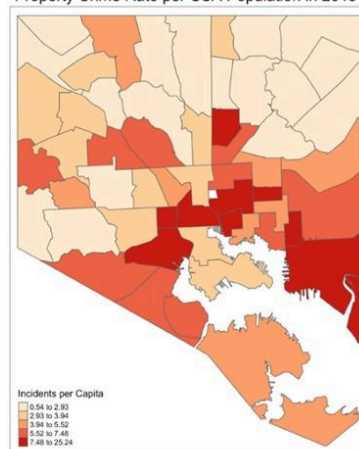
Property Crime Rate per CSA Population in 2016



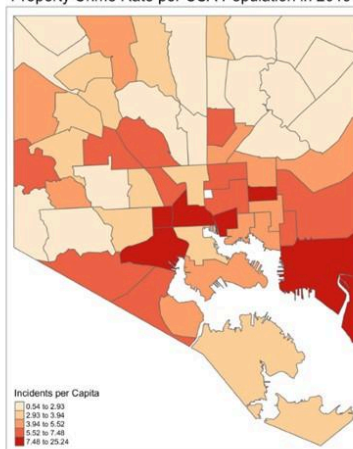
Property Crime Rate per CSA Population in 2017



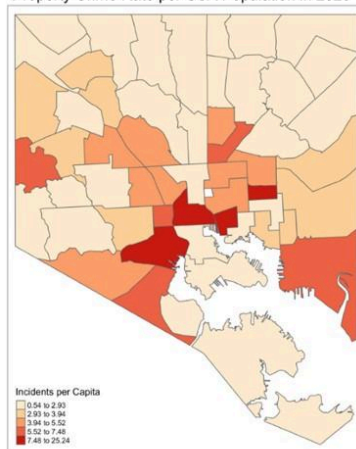
Property Crime Rate per CSA Population in 2018



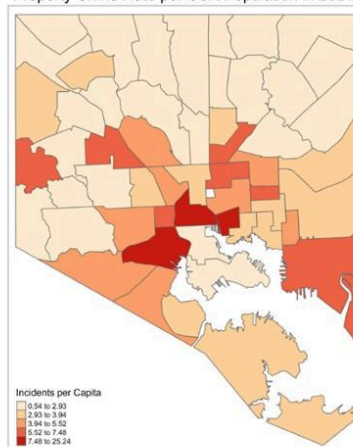
Property Crime Rate per CSA Population in 2019



Property Crime Rate per CSA Population in 2020

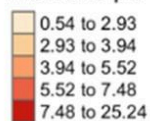


Property Crime Rate per CSA Population in 2021



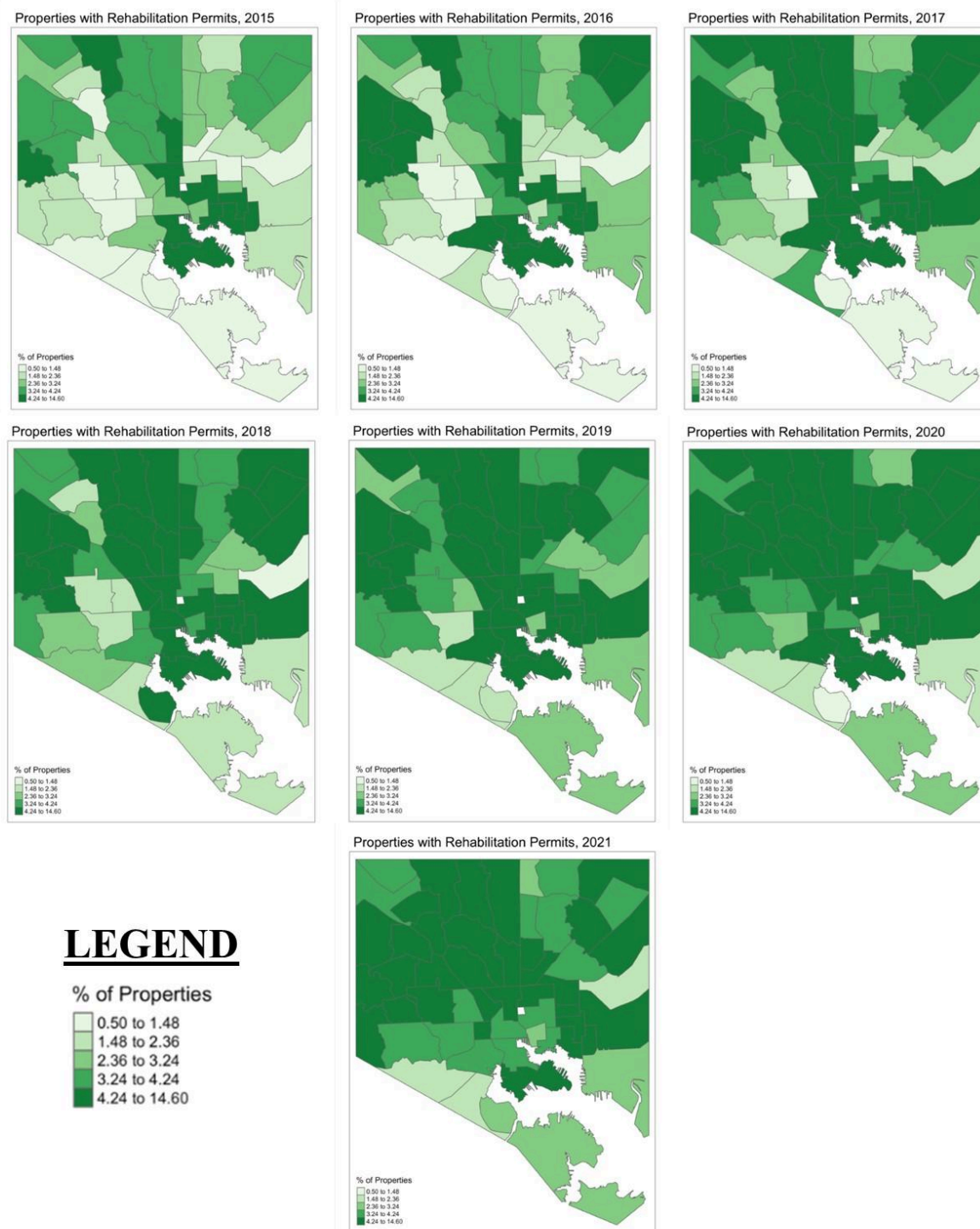
LEGEND

Incidents per Capita



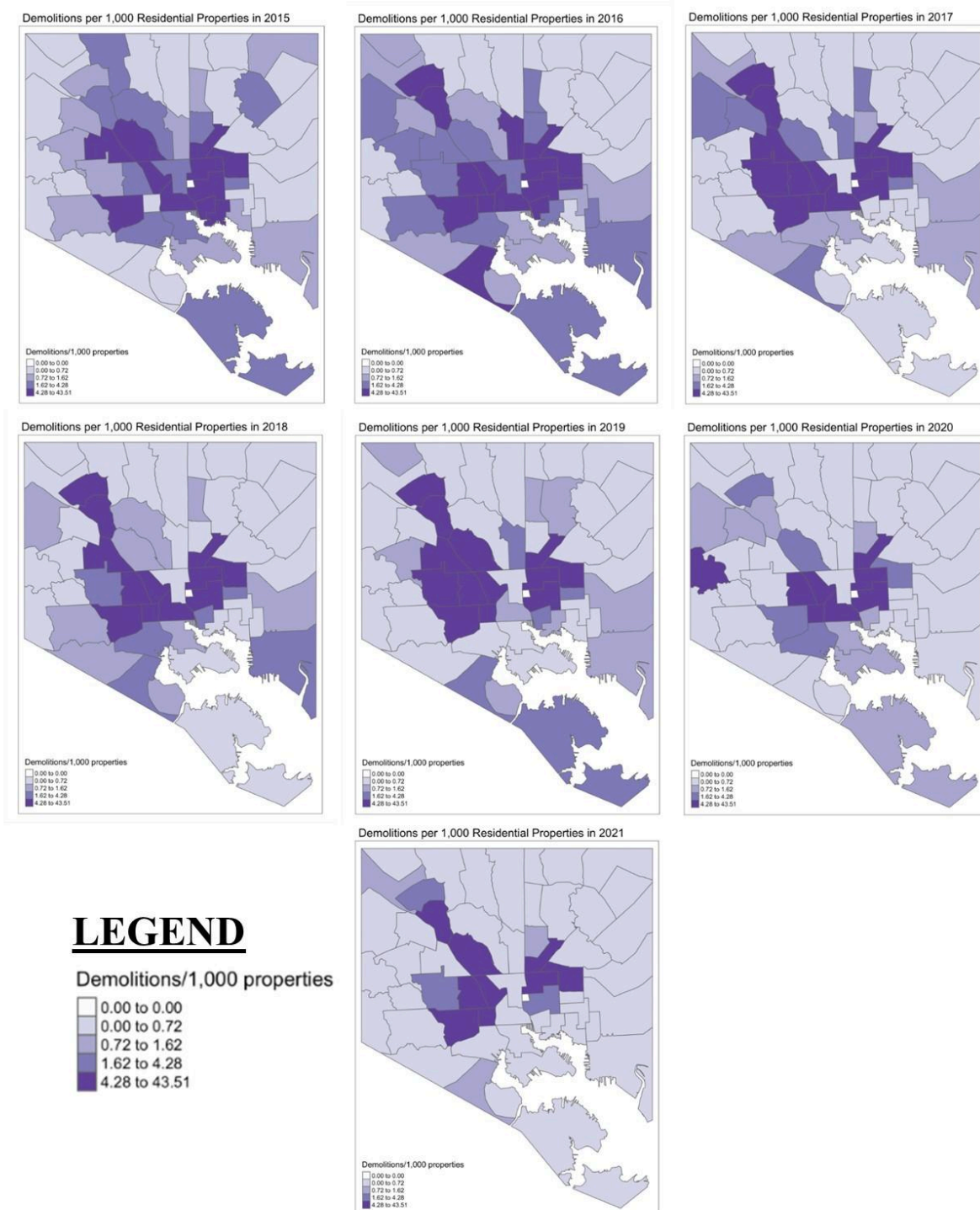
F.4. Prevalence of Rehabilitation Permits

APPENDIX F.4: PROPERTIES WITH REHABILITATION PERMITS



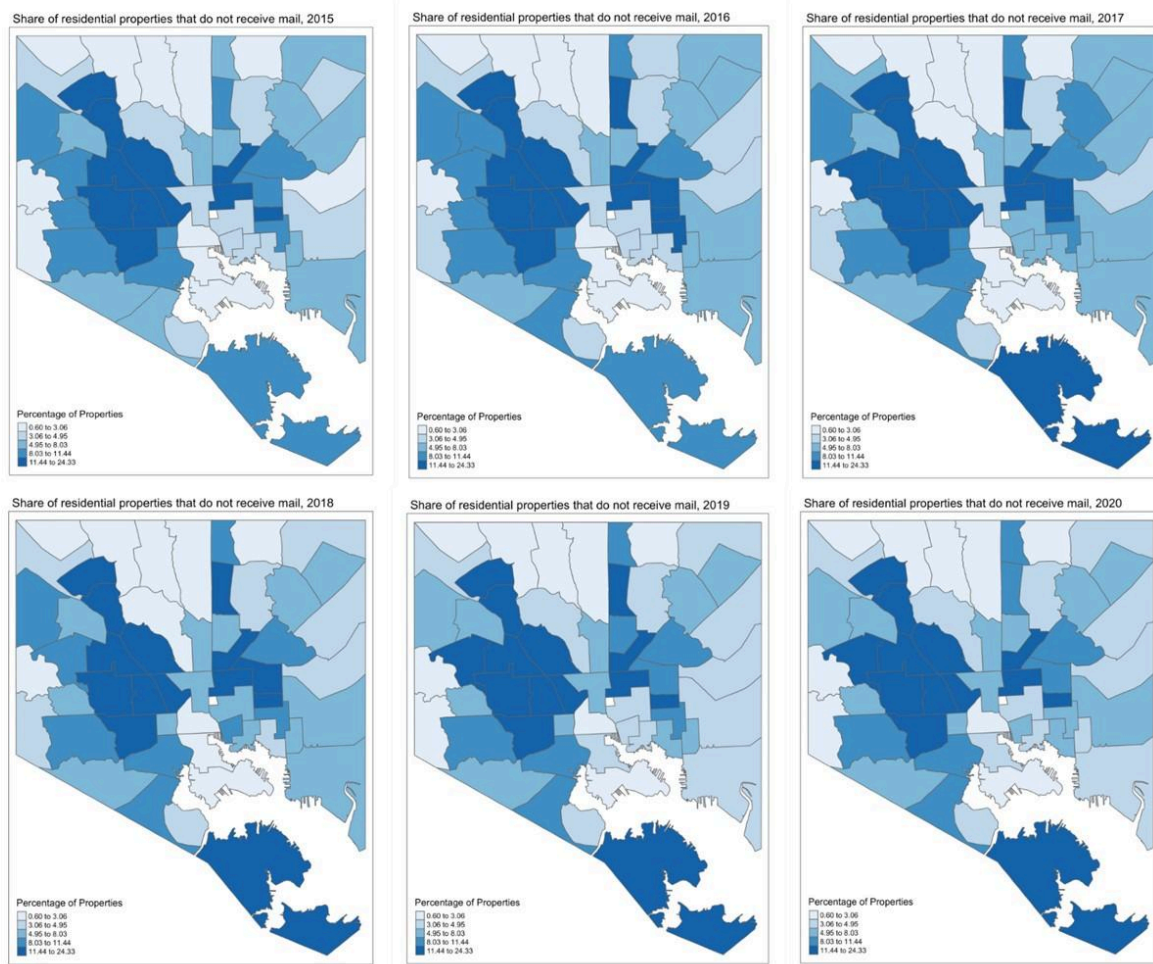
F.5. Prevalence of Demolition Permits

APPENDIX F.5: DEMOLITIONS PER 1,000 RESIDENTIAL PROPERTIES



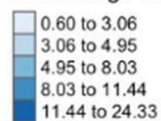
F.6. Share of Unoccupied Homes

APPENDIX F.6: SHARE OF RESIDENTIAL PROPERTIES THAT DO NOT RECEIVE MAIL



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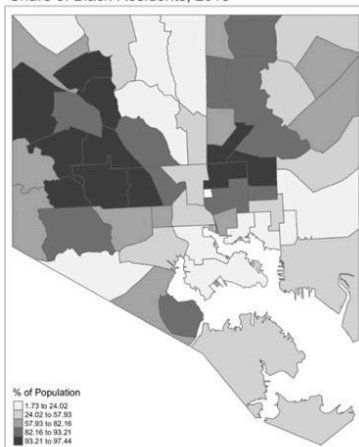
Percentage of Properties



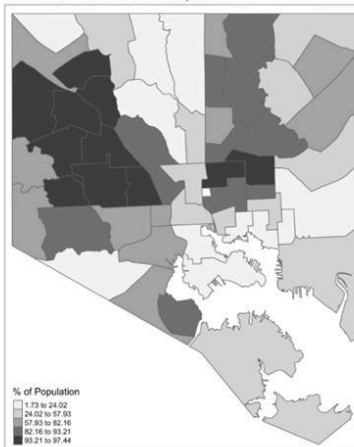
F.7. Share of Black Residents

APPENDIX F.7: SHARE OF BLACK RESIDENTS

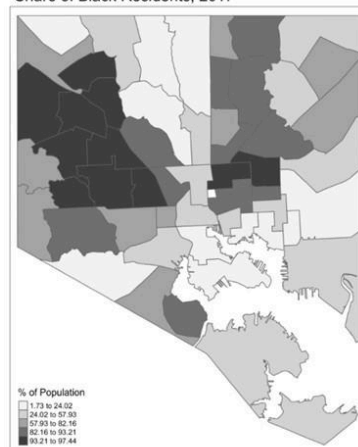
Share of Black Residents, 2015



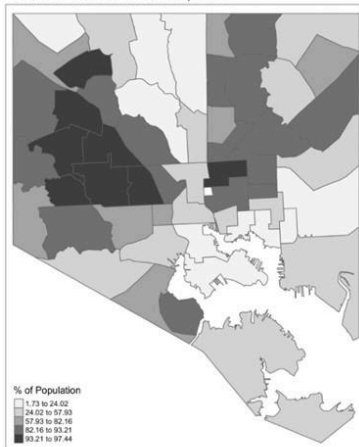
Share of Black Residents, 2016



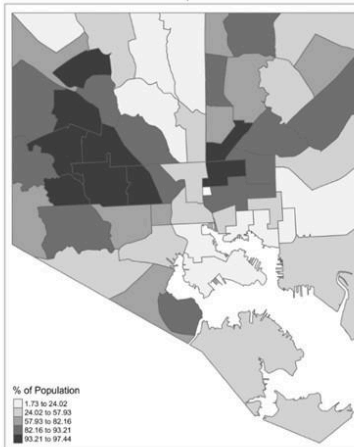
Share of Black Residents, 2017



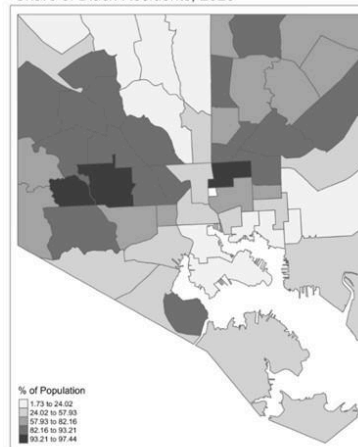
Share of Black Residents, 2018



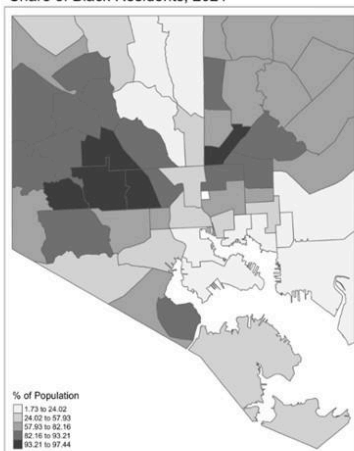
Share of Black Residents, 2019



Share of Black Residents, 2020



Share of Black Residents, 2021



LEGEND

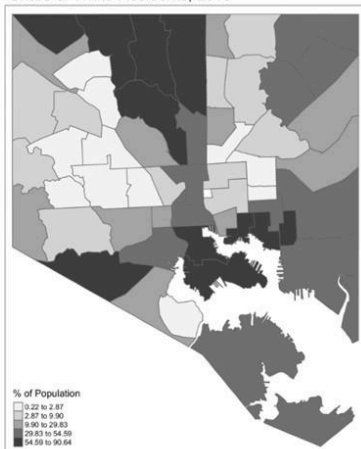
% of Population

1.73 to 24.02
24.02 to 57.93
57.93 to 82.16
82.16 to 93.21
93.21 to 97.44

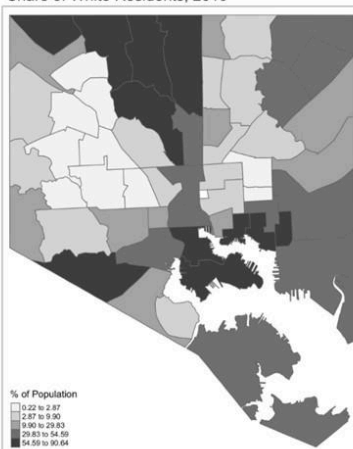
F.8. Share of White Residents

APPENDIX F.8: SHARE OF WHITE RESIDENTS

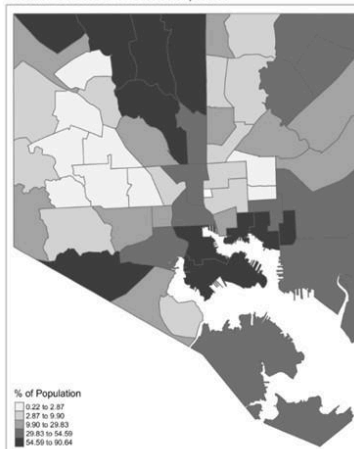
Share of White Residents, 2015



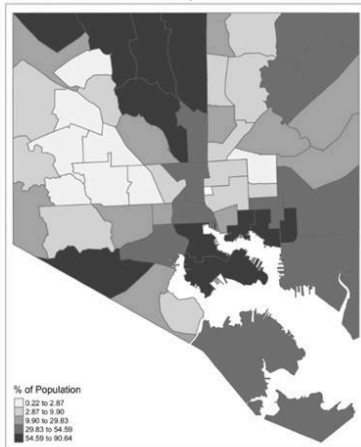
Share of White Residents, 2016



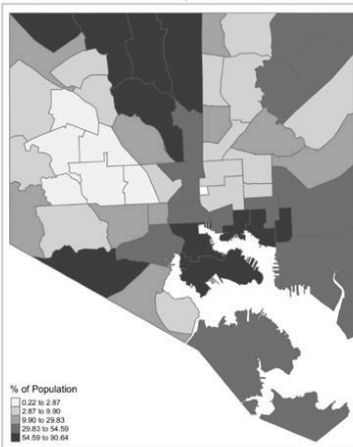
Share of White Residents, 2017



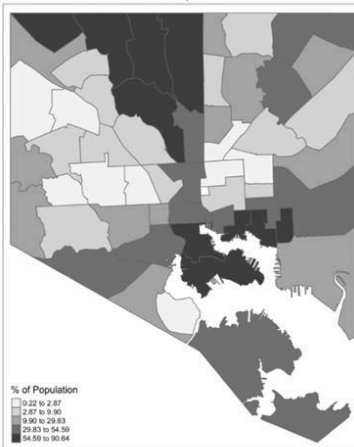
Share of White Residents, 2018



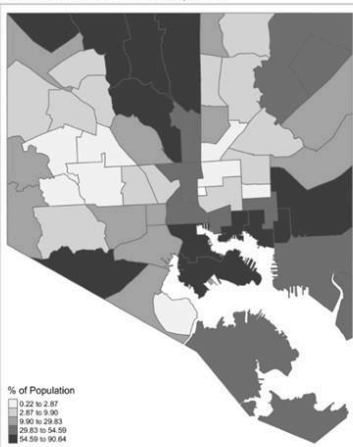
Share of White Residents, 2019



Share of White Residents, 2020

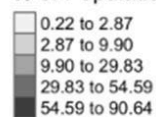


Share of White Residents, 2021



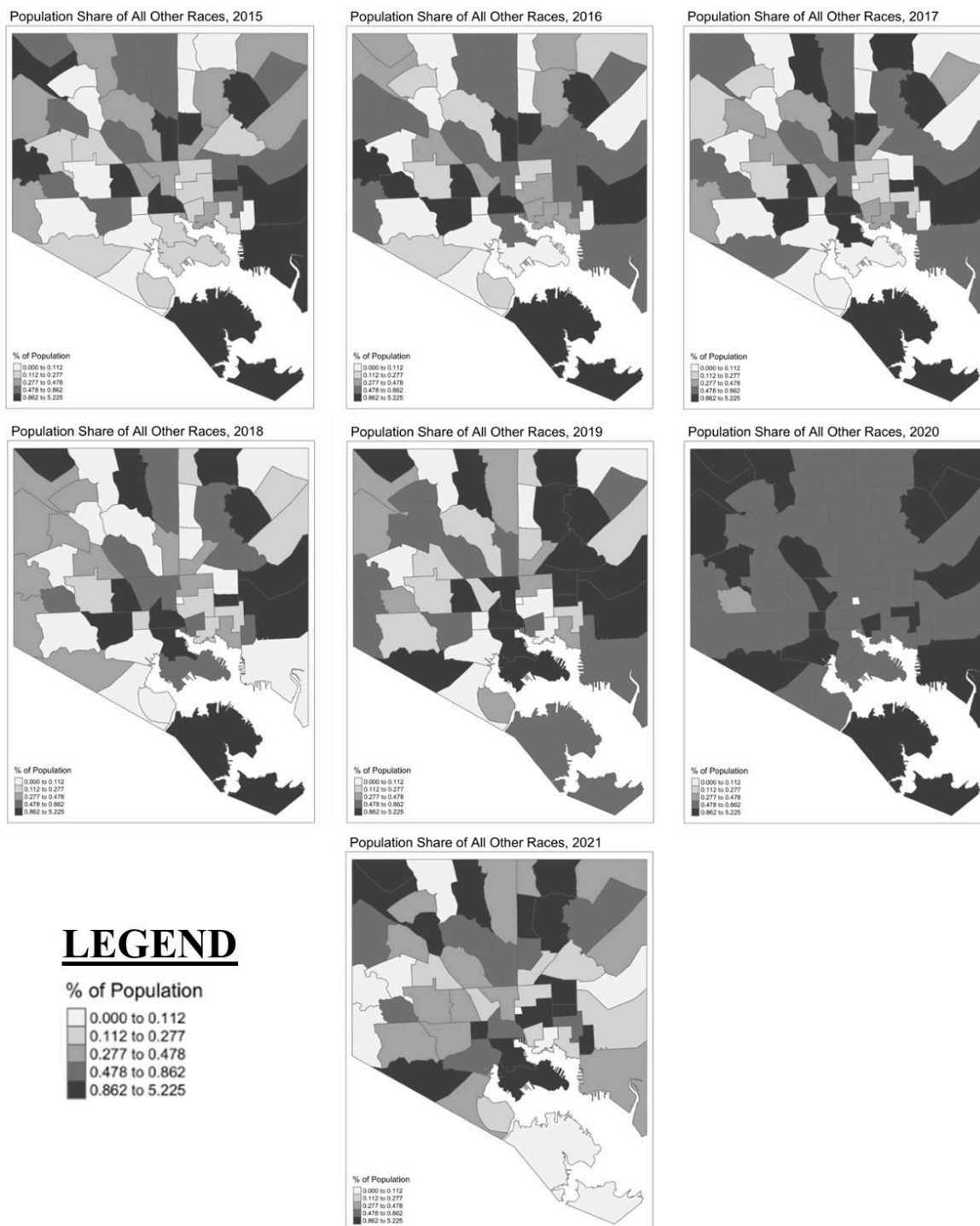
LEGEND

% of Population



F.9. Share of Residents of All Other Races

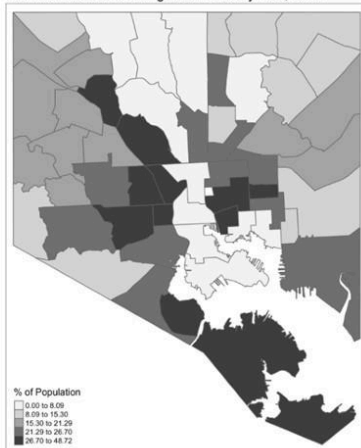
APPENDIX F.9: POPULATION SHARE OF ALL OTHER RACES



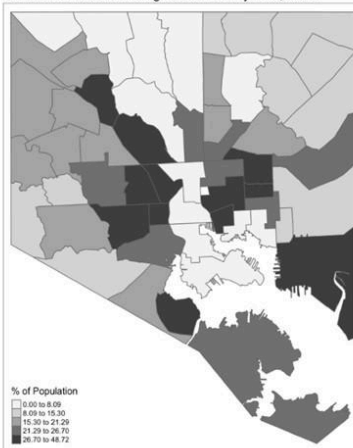
F.10. Share of Residents Below the Poverty Line

APPENDIX F.10: SHARE OF RESIDENTS LIVING BELOW THE POVERTY LINE

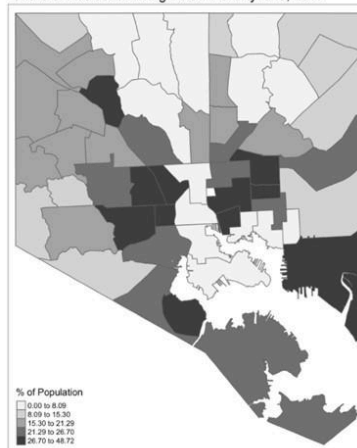
Share of Residents Living Below Poverty Line, 2015



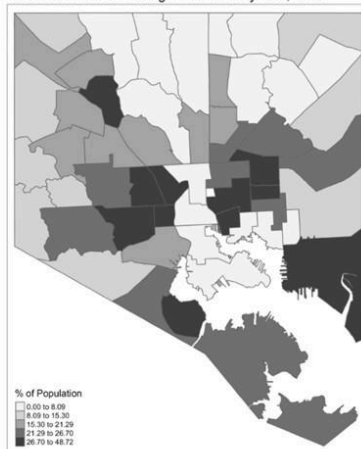
Share of Residents Living Below Poverty Line, 2016



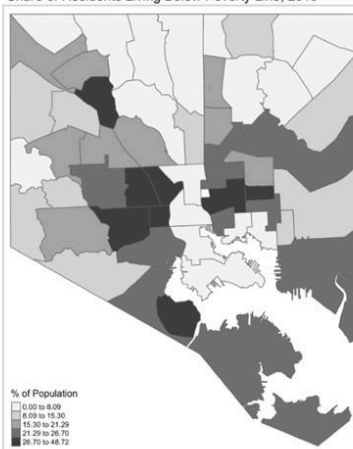
Share of Residents Living Below Poverty Line, 2017



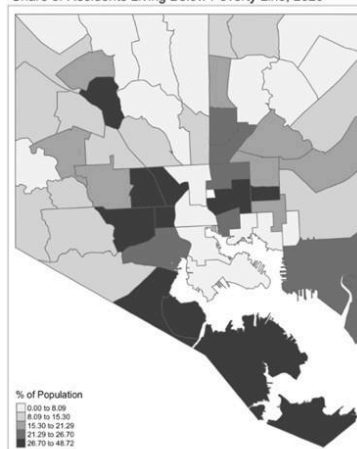
Share of Residents Living Below Poverty Line, 2018



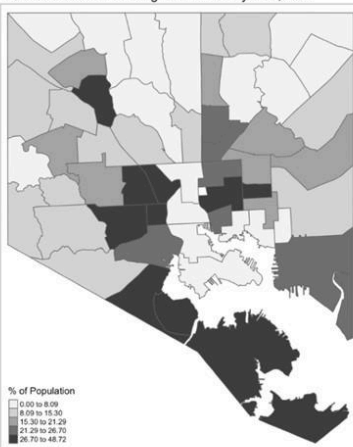
Share of Residents Living Below Poverty Line, 2019



Share of Residents Living Below Poverty Line, 2020



Share of Residents Living Below Poverty Line, 2021



LEGEND

% of Population

